Behavioral Responses to Air Quality Information: Implications for Optimal Policy

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Abstract

Air quality alerts can improve welfare by helping individuals avoid pollution exposure, but frequent warnings may reduce responsiveness and weaken the effectiveness of the alert system. This paper studies how people's outdoor activity responds to air quality alerts, how that response varies with recent alert history, and how these behavioral dynamics affect the optimal alert threshold. A model of optimal alert policy is combined with a fuzzy regression discontinuity design using air quality forecasts, alert issuances, and high-frequency foot traffic data from six U.S. metropolitan areas. Near the alert threshold, receiving an alert reduces daily visits to outdoor recreational points of interest by approximately 3.3 visits, or 9 percent relative to the sample mean. The response varies substantially with recent alert history: alerts reduce visits by roughly eight visits when one or two alerts have occurred during the previous seven days, but have little detectable effect after three or more recent alerts. A counterfactual simulation calibrated to New York City-specific pollution dynamics and behavioral responses places the welfare-maximizing alert threshold at approximately 80, below the current AQI threshold of 101. The results show that optimal warning policy depends not only on the pollution risks communicated by an alert, but also on how the frequency and sequencing of repeated warnings affect behavioral responsiveness.

1 Introduction

Air pollution imposes substantial costs on health and economic activity,[1] but individuals can reduce some of these costs through avoidance behavior. They may spend less time outdoors, shift travel patterns, or invest in defensive goods such as indoor air filtration (Neidell, 2009; Graff Zivin and Neidell, 2009; Cutter and Neidell, 2009; Deschênes et al., 2017; Ito and Zhang, 2020). The extent to which such avoidance occurs depends on information about air pollution. Real-time air quality information is increasingly available via mobile apps, websites, and other media, but availability alone does not ensure that information is salient, understood, or acted upon. Individuals may ignore pollution information, misunderstand how pollution values translate into health risks, or fail to adjust behavior when avoidance is costly. As emphasized by Greenstone and Jack (2015), revealed defensive behavior may therefore fall short of the privately optimal level when individuals are imperfectly informed, misunderstand the damages associated with pollution, or face other behavioral frictions. Consistent with this view, prior work shows that households value clean air, and that willingness to pay changes when individuals receive new information about pollution (Deschênes et al., 2017; Ito and Zhang, 2020; Barwick et al., 2024).

Public air quality information is designed to address this problem. In the United States, air pollution information is communicated primarily through the air quality index (AQI) and air quality alerts. The AQI maps pollutant concentrations onto a standardized index, and alerts are issued when pollution is forecasted to meet or exceed thresholds associated with unhealthy conditions. The primary threshold corresponds to an AQI of 101. While this threshold may be appropriate for conveying the health risks associated with pollution, it is not necessarily the threshold that maximizes welfare. Alerts matter in that they change beliefs and behavior, and their welfare value depends on whether individuals notice them, understand them, and adjust their exposure in response.

This creates a tradeoff in alert policy design. Lowering the alert threshold provides information on more days and may encourage avoidance when pollution is moderately elevated. At the same time, lowering the threshold changes the way recipients experience alerts: warnings become more frequent and are more likely to arrive after other recent warnings. Because the effect of an alert may depend on this recent history, changing the threshold can alter not only the number of warnings issued, but also the behavioral response to those warnings. Responsiveness may strengthen over some range of recent alert exposure and weaken over another. The welfare-maximizing threshold must therefore balance the informational gains from issuing alerts more often against the effect of alert frequency on the effectiveness of subsequent warnings.

[1]For a review of the intersection of air quality and health outcomes, see Brewer et al. (2023); for a review of the intersection of air quality and non-health outcomes, see Aguilar-Gomez et al. (2022)

The central question in this paper asks how regulators should choose an optimal air quality alert threshold when the effectiveness of an alert depends on recent alert exposure. Answering this question requires evidence on two empirical inputs. First, do alerts induce avoidance behavior? Second, how does the behavioral response vary with recent alert history? I answer these questions by combining high-frequency foot traffic data with air quality forecasts and alerts. I estimate behavioral responses near the alert threshold, examine how these responses vary with recent alert history, and incorporate the resulting estimates into a model of optimal alert policy.

I first develop a theoretical model that seeks to understand how behavior affects optimal policy. In the model, people receive and respond to air quality alerts, but responsiveness can change when alerts are sufficiently frequent. The regulator chooses an alert threshold to minimize the welfare loss from misperceived pollution. Lower thresholds provide information on more days, but also change how alerts are distributed across recent-alert states with different levels of responsiveness, possibly shifting individuals into states where they are less responsive. The optimal threshold balances these competing forces.

I then investigate responsiveness to air quality alerts using a fuzzy regression discontinuity (RD) design. My primary data sources for this analysis are high-frequency foot traffic data from Advan Research,[2] which tracks the number of visits to various points-of-interest (POIs) across cities, and air quality forecast and alert data from the Environmental Protection Agency's AirNow archive. I find that POIs that just receive an alert experience about 3.3 fewer visits relative to those POIs that just miss an alert. Relative to a mean in my sample of about 36.8 visits, this constitutes about a 9 percent reduction in visits following an alert.

Next, I investigate whether responsiveness changes as a function of the alert history, by sequentially estimating my fuzzy RD model, each time conditioning my data on the number of recent alerts. I then plot the estimates from each fuzzy RD regression to see how responsiveness changes. I find that effect magnitudes indeed change in the number of recent alerts. People are unresponsive to alerts when there has been no recent alerts. However, when there have been 1 or 2 recent alerts, people respond strongly, leading to a reduction of about 8 visits on days that just receive an alert relative to days that just miss one. As the alert history builds up, people cease responding to additional alerts: I find no responses to alerts when there have been 3 or more recent alerts. This diminishing effect is consistent with the idea of alert fatigue, as people stop responding to alerts after they become sufficiently frequent.

Finally, I combine my empirical findings with my theoretical model to perform a counterfactual simulation using data from the New York City core-based statistical area (CBSA). Parameters are directly taken from the data or estimated, and then the model

[2]These data were obtained via Dewey Data. Advan Research is a spinoff of Safegraph.

is solved to determine the optimal threshold. My findings suggest that lowering the threshold from its current AQI of 101 to about 80 would be welfare-enhancing in NYC. This recommendation to decrease the threshold is qualitatively the same across different assumptions about behavioral responsiveness.

This paper contributes to three strands of literature. First, I contribute to the literature on environmental information and avoidance behavior by providing new causal evidence on how air quality alerts affect everyday outdoor activity. Second, I contribute to the emerging literature on dynamic responses to air quality alerts by flexibly estimating how responsiveness varies with recent alert exposure, without imposing monotonicity. Third, I contribute to the literature on the design of public warning systems by developing and estimating a framework in which the alert threshold determines not only how frequently warnings are issued, but also the recent-alert states in which they occur and the resulting effectiveness of the alert system. The framework also allows me to examine how pollution persistence and forecast accuracy shape the optimal threshold.

My first contribution is to the literature on environmental information provision and avoidance behavior. Existing work shows that individuals respond to air quality information across a range of settings. Neidell (2009) finds that ozone alerts in Los Angeles reduced hospital visits among children with asthma, while Cutter and Neidell (2009) show that air quality alerts in San Francisco impacted transportation behavior by reducing traffic volumes and increasing use of public transit. More recent work finds similar evidence in different contexts. Anderson et al. (2022) find that air quality alerts in South Korea reduced cardiovascular and respiratory healthcare expenditures, while Ito and Zhang (2020) show that willingness to pay for clean air increased after the U.S. Embassy began disclosing air quality information in China. Barwick et al. (2024) show that real-time air quality disclosure in China induced avoidance behavior and affected welfare. Taken together, these papers establish that public information about air quality can impact behavior, health, and welfare. I extend this literature by bringing new data to the study of air quality alerts and avoidance behavior. Much of the existing literature examines outcomes including hospital admissions, transportation choices, or expenditures. Instead, I use high-frequency foot traffic data matched to air quality forecasts and alerts. These data provide more expansive geographic coverage than many existing studies, allowing me to study avoidance behavior across many locations rather than within a single city or activity. They also allow me to measure contemporaneous avoidance behavior directly across a wide range of points of interest. This extends upon previous work because avoidance responses to alerts may occur through changes in everyday activity, not just through less direct channels like healthcare utilization or aggregate transportation data. The foot traffic data therefore provide a more granular view of how individuals respond to public information about pollution.

My second contribution is to the small but growing literature on dynamic responses to air quality alerts. Alerts are not one-time treatments: they are issued repeatedly, and their effectiveness may depend on the recent history of alerts. This feature is especially important because air pollution is persistent, so alerts tend to cluster. This paper is most closely related to Graff Zivin and Neidell (2009), who study smog alerts in Los Angeles and their effect on visits to the L.A. Zoo and the Griffith Observatory and provide evidence consistent with alert fatigue. Newer papers support this finding, including Saberian et al. (2017), who examine bicycle ridership following alerts in Australia, and Dangel and Goeschl (2025), who estimate how driving activity changes during alert periods in Germany. I build on these studies by determining how responsiveness varies with recent alert histories across a broader set of locations and activities. Rather than imposing that repeated alerts monotonically reduce responsiveness, my empirical design flexibly estimates the shape of the relationship between recent alert exposure and the response to an additional warning. I find responsiveness initially increases with recent exposure before eventually attenuating as alerts continue to accumulate. The later decline is consistent with alert fatigue, while the non-monotonic pattern suggests the effectiveness of repeated warnings cannot be summarized by a constant or uniformly declining response.

My third, and central, contribution is to connect these empirical dynamics to a framework for the optimal design of alert policy. To my knowledge, this is the first paper to develop and estimate a model of optimal air quality alert threshold choice that explicitly incorporates history-dependent responsiveness to repeated alerts. My model formalizes a tradeoff that is central to warning systems issuing repeating alerts: lowering the threshold provides information on more days, but it also changes the distribution of warnings across recent-alert states with different levels of responsiveness. The regulator's problem, then, is not simply to maximize the number of warnings or to disclose information whenever pollution is elevated. Instead, the optimal threshold balances the informational value of more frequent alerts against changes in the average effectiveness of the warning system induced by alert frequency and sequencing. This framework allows me to translate empirical estimates of alert responsiveness into counterfactual alert thresholds and welfare comparisons.

More broadly, the paper connects to the economics literature on information provision and repeated behavioral interventions. A large body of work studies how information changes behavior, including through reminders, social comparisons, and other low-cost behavioral interventions.[3] Within this literature, a central question is whether the effects of information persist, decay, or change with repeated exposure. Allcott and Rogers (2014) show that repeated home energy reports generate dynamic behavioral responses over time, with initial action and backsliding, persistent effects after reports stop, and incomplete habituation. Related work studies the persistence of social nudges and the mechanisms through which repeated interventions generate longer-run behavior change

[3]See also Allcott (2011); Ferraro and Price (2013); Karlan et al. (2016); Haaland et al. (2023)

(Brandon et al., 2026). My paper studies a related dynamic in the context of environmental warnings, where information delivery is triggered by the underlying state of the environment rather than sent at fixed or randomly assigned intervals, as in an experiment. This distinction is important because the policy problem differs from the standard repeated intervention setting. In many information provision experiments, the researcher or policymaker chooses when to send messages. In my setting, the timing of information is governed by an alert threshold applied to a persistent environmental process. Because air pollution is serially correlated, repeated warnings are not randomly spaced. The regulator therefore chooses not only how much information to provide, but also how often individuals are exposed to warnings in recent-alert states with different levels of responsiveness. This links history-dependent responses to repeated information directly to the optimal design of the warning threshold.

The remainder of this paper is organized as follows: In Section 2, I provide background information about air quality, the air quality index, and air quality forecasts and alerts. In Section 3, I set up, solve, and discuss the implications of my theoretical model. In Section 4, I describe the data I use in my empirical analysis. In Section 5, I outline my empirical strategy, and present my results in Section 6. Section 7 builds and executes my counterfactual simulation. Section 8 concludes.

2 Background on Air Pollution

In this section, I provide background on the air pollutants this paper focuses on, as well as information regarding air quality measurement, the air quality index (AQI), and air quality forecasts and alerts. Unless otherwise noted, this section draws on information from technical reports published by the EPA.[4]

2.1 Particulate and Ozone Pollution

Particulate matter (PM) is a mixture of particles and liquid droplets suspended in the air. PM can be classified by size, with $PM_{2.5}$ and PM_{10} being the most commonly measured sizes. $PM_{2.5}$ refers to particles with a diameter of 2.5 micrometers or smaller, while PM_{10} refers to particles with a diameter of 10 micrometers or smaller. $PM_{2.5}$ is considered more harmful to human health because these particles can penetrate deep into the lungs and even enter the bloodstream. PM can be made up of hundreds of different chemicals and come from a variety of origins. Construction sites, unpaved roads, smokestacks, and fires are all sources. Additionally, PM can be formed in the atmosphere from chemical reactions between other pollutants, such as sulfur dioxide and nitrogen oxides emitted

[4]See U.S. Environmental Protection Agency (2018) and U.S. Environmental Protection Agency (2024).

from power plants, industrial processes, and vehicles.

Exposure to PM can effect both the respiratory and cardiovascular systems and have been linked to a variety of health problems, including heart attacks, arrythmia, asthma, and decreased respiratory and lung function. Long-term exposure has been associated with the development of respiratory and cardiovascular diseases, nervous system and cognitive problems, cancers, and even premature death.

Ozone is a gas that occurs naturally in the Earth's atmosphere. Ozone is most commonly found in the stratosphere, where it forms the ozone layer that protects the Earth from harmful ultraviolet radiation. However, ozone can also be found at ground level, where it is considered a pollutant, and is the primary component of "smog". It is formed when sunlight reacts with volatile organic compounds (VOCs) and nitrogen oxides (NO_x) emitted by vehicles, power plants, and industrial facilities. Because of ozone's formation process, it is more prevalent in the summer months when sunlight is more abundant.

Ozone exposure can cause respiratory symptoms and inflammation, and can exacerbate existing respiratory conditions such as asthma. While people with pre-existing respiratory conditions are more vulnerable to the effects of ozone, healthy individuals can also experience adverse health effects from exposure. Long-term exposure to ozone is thought to be an important risk factor for the development of asthma.

2.2 The Air Quality Index (AQI)

The Air Quality Index (AQI) is measure developed by the U.S. Environmental Protection Agency (EPA) to communicate air pollution levels to the public. AQI values begin at zero and increase as pollution levels rise. AQI values correspond to six escalating intervals of health risk, ranging from "Good" to "Hazardous" (See Table 1 for a full list of AQI categories). These intervals are also color-coded and accompanied by statements describing the severity of the pollution, the most affected groups, and recommendations for reducing exposure.

At any given time, there are potentially six different calculated AQIs, one for each of the pollutants regulated by the Clean Air Act: particulate matter ($PM_{2.5}$ and PM_{10}), ozone, lead, carbon monoxide, sulfur dioxide, and nitrogen dioxide. The highest of these six measures is then taken as the primary AQI for that location on that day. In the U.S., the highest AQIs are generally those for $PM_{2.5}$ and ozone (U.S. Environmental Protection Agency, 2006). These pollutants are measured in micrograms per cubic meter ($\mu g/m^3$) for $PM_{2.5}$ and parts per million (ppm) for ozone. AQI values are then calculated from raw pollutant concentrations.[5] The AQI is then rounded to the nearest integer.[6] The breakpoints for $PM_{2.5}$ and ozone are shown in Table 1.

Table 1: Breakdown of AQI Categories

AQI	$PM_{2.5}$ (µg/m^3)	O_3 (ppm)	Category	Color
0 – 50	0.0 – 9.0	0.000 – 0.054	Good	Green
51 – 100	9.1 – 35.4	0.055 – 0.070	Moderate	Yellow
101 – 150	35.5 – 55.4	0.071 – 0.085	Unhealthy for Sensitive Groups	Orange
151 – 200	55.5 – 125.4	0.086 – 0.105	Unhealthy	Red
201 – 300	125.5 – 225.4	0.106 – 0.200	Very Unhealthy	Purple
301+	225+	0.201+	Hazardous	Maroon

Notes: The dashed red line marks the AQI 101 threshold relevant for this paper, separating the "Moderate" category from the "Unhealthy for Sensitive Groups" category. The concentration breakpoints reported in this table reflect the updates the EPA made to the AQI in May of 2024. Prior to these changes, the "Good" category for $PM_{2.5}$ was 0.0 – 12.0 μg/m^3 and the "Moderate" category was 12.1 – 35.4 μg/m^3.

Each pollutant has a national ambiant air quality standard (NAAQS) determined by the EPA. The AQI for each pollutant combines the NAAQS with prevailing health research to determine breakpoints and the shape of the AQI schedule. An AQI of 100 corresponds to the short-term NAAQS for that pollutant, with values above 100 indicating unsafe levels of pollution, and values below 100 indicating satisfactory air quality.

Metropolitan Statistical Areas (MSAs) with a population of at least 350,000 are required to record and report daily AQI values to the public. Any pollutant for which the AQI stays below 50 for a year does not need to be reported, but reporting for that pollutant must resume if its AQI rises above 50 in subsequent years.

MSAs are required to report the reporting area and period, the primary pollutant, the AQI value, the category and corresponding color, and the sensitive groups for all pollutants with an AQI above 100. In addition, many agencies can also choose to report any or all of the following: AQI forecasts, hourly concentrations, health effects and accompanying cautionary statements, potential causes for unusual AQI values, inter-reporting area AQI values, raw pollutant concentrations, and AQIs for the non-primary pollutants.

[5] Pollution concentrations are mapped to AQI values using the following formula:

$$I_p = \frac{I_{\text{Hi}} - I_{\text{Lo}}}{\text{BP}_{\text{Hi}} - \text{BP}_{\text{Lo}}}(C_p - \text{BP}_{\text{Lo}}) + I_{\text{Lo}},$$

where I_p is the AQI for pollutant p, C_p is the rounded concentration of pollutant p, BP_{Lo} and BP_{Hi} are the lower and upper concentration breakpoints for pollutant p, and I_{Lo} and I_{Hi} are the AQI values corresponding to the lower and upper breakpoints.

[6] The use of a discrete running variable can create problems for regression discontinuity designs (See Lee and Card (2008)). Where possible, I use an unrounded AQI value as the running variable instead.

2.3 Air Quality Forecasts and Alerts

Air quality alerts are issued based on forecasted rather than realized AQI. In most jurisdictions, an alert is nominally triggered when forecasted AQI exceeds 100. In practice, however, both components of this process are imperfect: forecasts can differ substantially from realized pollution, and agencies do not issue alerts deterministically at the stated cutoff. These features are central to my empirical design because they make forecasted AQI the relevant running variable and imply a fuzzy rather than sharp discontinuity in alert issuance.

AQI forecasts are predictions of the AQI for subsequent days, predominantly the next day. These forecasts are typically performed by state or local forecasters. Different forecasters use different methods to produce their forecasts, but they broadly rely on a combination of historical data, meteorological information, and atmospheric models.

Many agencies also issue air quality alerts, or air quality action days, when the AQI is forecasted to exceed a certain threshold, typically 100. Issuances of these alerts are intended to inform the public about the expected air quality and to encourage people to take precautions to reduce their exposure to air pollution. Alerts are generally issued by state or local agencies, and the precise conditions under which alerts are issued vary by agency. Alerts are disseminated through a variety of channels, including television, radio, social media, and mobile applications. Some agencies also provide alerts via text message or email.

While the official alert trigger across nearly all jurisdictions that issue alerts is a forecasted AQI exceeding 100, evidence shows forecasting and issuing alerts is not a perfect process. Figure 1 shows the relationship between forecasted AQI and actual AQI, with the color of each grid cell indicating how likely that forecasted-actual AQI pair is to receive an alert. The dotted blue 45-degree line represents perfect forecast accuracy. If forecasts perfectly predicted realized AQI, all observations would lie along this line. Instead, observations are dispersed around it, in some cases substantially, indicating considerable forecast error. More observations also fall below the line than above it, meaning that forecasts tend to overstate realized AQI.

The figure also shows that alerts are not issued deterministically at the official threshold. The vertical dashed red line marks a forecasted AQI of 101, the first integer value above 100. If agencies followed their stated policies perfectly, observations to the right of this line would have an alert probability of 100 percent, while those to the left would have an alert probability of zero. Instead, some forecast values above the threshold do not receive alerts, while some values below it do.

These facts have two implications for my RD empirical design. First, because alerts are issued based on forecasts rather than realized AQI, I use the forecast as the running variable. Second, non-deterministic alert issuance at the threshold implies a fuzzy (rather

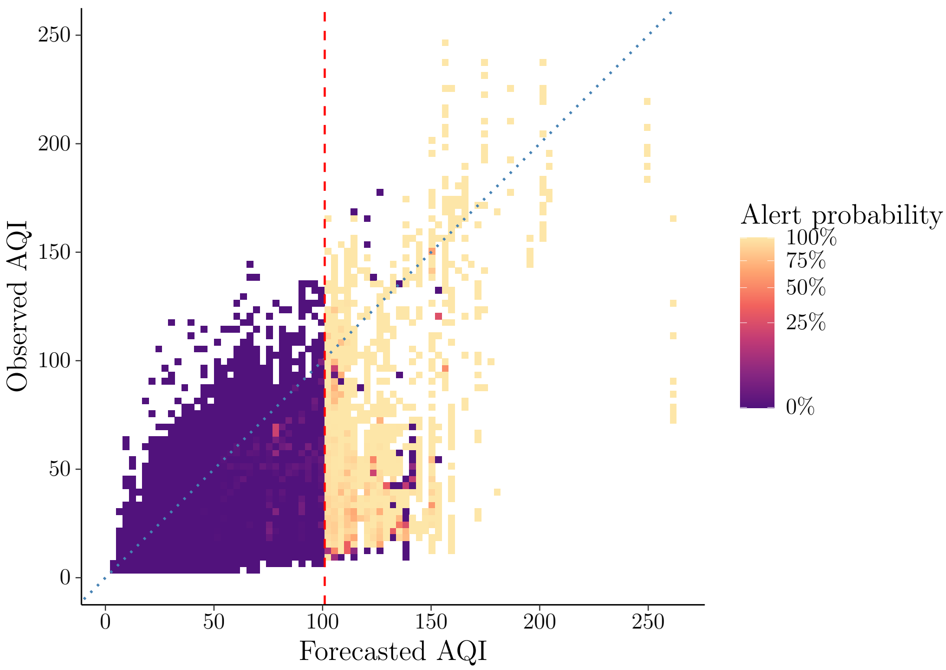


Figure 1: Forecast and Alert Accuracy

Notes: This figure compares forecasted AQI with realized AQI in the analysis sample. Each grid cell represents a forecasted-realized AQI combination, and its color reports the share of observations in that cell for which an air quality alert was issued. The dotted blue 45-degree line denotes perfect forecast accuracy; observations below the line correspond to forecasts that overstate realized AQI, while observations above the line correspond to forecasts that understate realized AQI. The dashed red vertical line marks the official alert threshold at a forecasted AQI of 101. Cells with no observations are omitted.

than sharp) discontinuity. Accordingly, when examining the effect of alerts, I use crossing a forecasted AQI of 101 as an instrument for receiving an alert.

3 Theoretical Framework

This section develops a model of optimal air quality alert policy when individuals may become less responsive as alerts become more frequent. A regulator chooses an alert threshold, and a consumer adjusts their behavior based on their perceived pollution exposure, which depends on whether an alert is issued and on how much weight they place on that signal. The model reveals a central tradeoff: lowering the threshold improves information by warning people on more days, but it can also reduce the effectiveness of each alert if repeated alerts generate fatigue. The optimal threshold therefore balances alert frequency against alert effectiveness. I first characterize this tradeoff in an i.i.d. setting, then extend the model to allow serial correlation in pollution, showing that persistence matters as clustered alerts can further weaken responses over time.

3.1 Setup

Consider a calendar of days indexed by t, where $t = 1, 2, 3, ...$ Each day has its own draw of pollution, ϕ_t, where $\{\phi_t\}$ is an independent and identically distributed sequence drawn from some distribution F with density f. Pollution is normalized to lie on $[0, 1]$, with 0 meaning there is no pollution and 1 meaning pollution is at its maximum value. Suppose there is a regulator who issues alerts about air quality based on an alert threshold $A \in (0, 1]$. If day t's pollution value meets or exceeds the threshold, that is, if $\phi_t \geq A$, then an alert is issued. Otherwise, if $\phi_t < A$, no alert is issued. Because ϕ_t has cdf F, then $Pr(\phi_t \geq A) = 1 - F(A)$ is the probability a day receives an alert and $\Pr(\phi_t < A) = F(A)$ is the probability otherwise.

It is possible that the impact of an alert depends on the history of recent alerts. Define a state variable n_t that tracks the number of alerts issued in the previous ρ days. Specifically,

$$n_t := \sum_{i=1}^{\rho} \mathbb{1}\{\phi_{t-i} \geq A\}, \tag{1}$$

where ρ is the length of the recent-alert window, and thus $n_t \in \{0, 1, \ldots, \rho\}$.[7] The variable n_t therefore summarizes the recent sequence of alerts faced prior to day t.

[7]In general, ρ is a parameter that can be estimated. For simplicity, I fix ρ at 7 in the empirical portions in later sections.

3.2 Consumer Beliefs and Responsiveness

Suppose a representative consumer has some true utility function that takes the form

$$U^{\text{TRUE}}(x,\phi) = u(x) - h(x,\phi) = bx - \frac{1}{2}x^2 - \phi x.$$

Here, $u(x) = bx - \frac{1}{2}x^2$ is the consumer's utility from allocating time x towards some activity that would expose them to pollution, and $h(x,\phi) = \phi x$ is the pollution damage the consumer receives from their choice of x. An optimizing consumer will choose x such that the marginal benefit of x is equal to its marginal cost. That is, they will choose x such that

$$\frac{du(x)}{dx} = \frac{\partial h(x,\phi)}{\partial x} \Rightarrow x^* = b - \phi.$$

Call x^* the optimal choice of x under U^{TRUE} with pollution level ϕ.

Now, suppose that people are misinformed about the level of air pollution. Define $\tilde{\phi}_1$ as the consumer's posterior perceived belief about the level of pollution. Then, *perceived* consumer utility is reflected by

$$U^{\text{PERC}}(x,\tilde{\phi}_1) = u(x) - h(x,\tilde{\phi}_1) = bx - \frac{1}{2}x^2 - \tilde{\phi}_1 x.$$

The misinformed consumer will choose x such that the marginal benefit of x is equal to its perceived marginal cost:

$$\frac{du(x)}{dx} = \frac{\partial h(x,\tilde{\phi}_1)}{\partial x} \Rightarrow \tilde{x} = b - \tilde{\phi}_1.$$

Call $\tilde{x}$ the optimal choice of x under U^{PERC} with perceived pollution level $\tilde{\phi}_1$.

The consumer's belief about air pollution depends on whether they received an alert. In the absence of an alert, assume the consumer believes pollution to be at the average level of pollution below the alert threshold A. Formally, the consumer's prior perceived level of pollution is

$$\tilde{\phi}_0(A) := \mathbb{E}[\phi_t \mid \phi_t < A] = \frac{\int_0^A \phi f(\phi) d\phi}{F(A)}. \tag{2}$$

An implication of this parameterization is that the absence of an alert does convey some signal to consumer about the severity of pollution on day t. The consumer's prior will increase in A according to the shape of distribution of ϕ.

The consumer's posterior perceived pollution level on day t, $\tilde{\phi}_{1,t}$, can then be split into a case where they receive an alert and a case where they do not. Specifically,

$$\tilde{\phi}_{1,t} = \begin{cases} (1-\alpha(n_t))\tilde{\phi}_0(A) + \alpha(n_t)\phi_t & \text{if } \phi_t \geq A, \\ \tilde{\phi}_0(A) & \text{if } \phi_t < A. \end{cases} \tag{3}$$

Beginning with the second case, if ϕ_t is less than the alert threshold A, then no alert is issued, and the consumer's posterior belief about pollution is equal to their prior belief about pollution. As defined in Equation 2, the consumer's prior belief is equal to the expected value of ϕ_t conditional on being below the alert threshold.

If ϕ_t meets or exceeds the alert threshold and an alert is issued, the consumer's posterior belief is equivalent to a weighted average of their prior belief, $\tilde{\phi}_0(A)$, and the true pollution signal, ϕ_t. The weights are governed by the consumer's behavioral response function, $\alpha(n_t)$, which captures how strongly the consumer updates after receiving an alert, conditional on the recent history of alerts. Let

$$\alpha : \{0, 1, \ldots, \rho\} \to [0, 1].$$

When $\alpha(n_t)$ is relatively large for a particular day t, then more weight will be placed on the true signal, and the consumer's posterior belief will end up closer to ϕ_t. If $\alpha(n_t)$ is relatively small for a particular day t, then more weight will be placed on the consumer's prior belief, and the consumer's posterior belief will end up further from ϕ_t.

The formulation above leaves the shape of $\alpha(\cdot)$ unrestricted. This flexibility is important because the representative consumer's responsiveness to an alert may reflect two countervailing behavioral channels. A fatigue channel may reduce responsiveness as recent alerts become more frequent. The representative consumer may become accustomed to warnings, may find repeated avoidance costly, or may place less weight on additional alerts after having observed alerts recently. This channel tends to lower $\alpha(n)$ as n increases. At the same time, a salience channel may increase responsiveness. Recent alerts may raise the representative consumer's attention to air quality risk, make pollution information easier to recall, or coincide with other signals that make the underlying risk harder to ignore. This channel tends to raise $\alpha(n)$ as n increases.

Resultingly, the net shape of $\alpha(\cdot)$ is theoretically ambiguous. I therefore treat $\alpha(\cdot)$ as an object to be estimated rather than impose monotonicity or a specific functional form.

A key assumption I make is that $\alpha(\cdot)$ is independent of the alert threshold A. Changing A changes the distribution of alert days across recent-alert states n_t, but does not alter the responsiveness associated with each state. Thus, the threshold affects the allocation of alert days across the response function, not the shape of the function itself.[8]

[8]It is possible this is an incorrect assumption and $\alpha(\cdot)$ *is* a function of A, but in practice, this is a difficult thing to determine without variation in A, which I do not have in my setting.

3.3 Regulator's Problem

The regulator seeks to choose A such that the difference between true utility evaluated at x^* and true utility evaluated at $\tilde{x}$ is minimized. Call this difference the decision wedge $\Delta(\phi, \tilde{\phi}_1)$. Formally,

$$\Delta(\phi, \tilde{\phi}_1) = U^{\text{TRUE}}(x^*(\phi), \phi) - U^{\text{TRUE}}(\tilde{x}(\tilde{\phi}_1), \phi).$$

Because U^{TRUE} is quadratic in x, the decision wedge can be written as

$$\Delta(\phi, \tilde{\phi}_1) = \frac{1}{2}(\phi - \tilde{\phi}_1)^2$$

by substituting in x^* and $\tilde{x}$ and simplifying.[9] Thus, the welfare loss from misperceiving pollution is proportional to the squared difference between true and perceived pollution. The regulator can then maximize consumer utility by minimizing this difference without having to know the consumer's underlying utility function.[10]

Using the law of total expectation and assuming stationarity, the expected decision wedge can be written as a probability-weighted average of the decision wedge below and above the alert threshold A, as derived in Appendix A.1.2:

$$\mathbb{E}[\Delta] = \Pr(\phi < A)\mathbb{E}[\delta_0(\phi) \mid \phi < A] + \Pr(\phi \geq A)\mathbb{E}[\delta_1(\phi, n) \mid \phi \geq A]. \tag{4}$$

The threshold A affects this object both by changing the probability of each case and by changing the wedge realized within each case.

On alert days, responsiveness to alerts depends on the recent-alert state n. For a given threshold A, define weights

$$w_j(A, \rho) := \Pr(n = j \mid \phi \geq A), \qquad j = 0, 1, \ldots, \rho.$$

That is, $w_j(A, \rho)$ is the proportion of alert days that occur when there have been j alerts in the previous ρ days. Then,

$$\mathbb{E}[\delta_1(\phi, n) \mid \phi \geq A] = \sum_{j=0}^{\rho} w_j(A, \rho)\, \mathbb{E}[\delta_1(\phi, j) \mid \phi \geq A, n = j].$$

Among alert days, the expected wedge depends on how alert days are distributed across recent-alert states. The overall expected alert-day wedge is therefore the average of the wedge in each state j, weighted by the share of alert days that occur in that state.

[9] See Appendix A.1.1 for derivation.

[10] This expression holds exactly under the assumed functional form for utility specified above. With a more general smooth utility function, it can instead be interpreted as a second-order Taylor approximation of the welfare loss around the consumer's true optimal choice, $x^*(\phi)$.

After making the appropriate substitutions and simplifying,[11] Equation 4 can be rewritten as

$$\mathbb{E}[\Delta] = \frac{1}{2}\Big[\int_0^A \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi \\ + \sum_{j=0}^{\rho} w_j(A,\rho)(1-\alpha(j))^2 \int_A^1 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi\Big].$$

Now, define

$$\mathcal{I}_0(A) := \int_0^A \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi, \tag{5}$$

$$\mathcal{I}_1(A) := \int_A^1 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi, \tag{6}$$

and define the average behavioral response attenuation factor as

$$W(A,\rho) := \sum_{j=0}^{\rho} w_j(A,\rho)(1-\alpha(j))^2. \tag{7}$$

The term $\mathcal{I}_0(A)$ measures the total expected decision wedge on non-alert days, while $\mathcal{I}_1(A)$ measures the corresponding baseline wedge on alert days. The object $W(A,\rho)$ captures how consumer responsiveness attenuates the alert-day wedge on average, weighting each state-specific attenuation factor $(1-\alpha(j))^2$ by the share of alert days that occur in recent-alert state j.

Thus, the regulator's problem is to choose A such that $\mathbb{E}[\Delta]$ is minimized:

$$\min_A \ \mathbb{E}[\Delta] = \frac{1}{2}[\mathcal{I}_0(A) + W(A,\rho)\mathcal{I}_1(A)]. \tag{8}$$

3.4 Optimal Alert Threshold

Taking the derivate of Equation 8 with respect to A and setting this equal to zero yields the first-order condition

$$0 = \mathcal{I}_0'(A) + W(A,\rho)\mathcal{I}_1'(A) + W'(A,\rho)\mathcal{I}_1(A). \tag{9}$$

The first two terms, $\mathcal{I}_0'(A) + W(A,\rho)\mathcal{I}_1'(A)$, capture the direct effects of changing the threshold, holding fixed changes in the average responsiveness of alerts. Increasing A expands the region below the cutoff where no alerts are issued while shrinking the region above the cutoff, which changes the expected decision wedge on each side of the cutoff. The term $\mathcal{I}_0'(A)$ is the marginal effect of this change on non-alert days, while

[11] See Appendix A.1.2.

$W(A,\rho)\mathcal{I}_1'(A)$ is the corresponding marginal effect on alert days, but scaled by the average attenuation factor $W(A,\rho)$.

The third term, $W'(A,\rho)\mathcal{I}_1(A)$, captures the indirect effect of changing the threshold through alert responsiveness. Because $\alpha(\cdot)$ is assumed to be independent of A, changes in $W(A,\rho)$ arise only through changes in the weights $w_j(A,\rho)$. That is, changes in average attenuation arise only through changes in how alert days are distributed across recent-alert states n. Thus, changes in A affect not only how many days receive alerts, but also how effective those alerts are on average. The importance of this margin is proportional to $\mathcal{I}_1(A)$, the amount of decision wedge that lies above the cutoff that can thus be affected by alerts.

The optimal threshold A^* balances these forces. Specifically, it trades off the net direct effect of reallocating days across the threshold against the indirect effect of changing the average effectiveness of alerts.

3.5 Relaxing Independence

In my baseline model, I assume pollution draws are i.i.d. In practice, air quality displays some degree of serial correlation, where ϕ_t depends on the draws of ϕ in previous days. Suppose now that $\{\phi_t\}$ is a stationary first-order Markov process on $[0,1]$ with transition kernel

$$G(\phi' \mid \phi) := \Pr(\phi_{t+1} \leq \phi' \mid \phi_t = \phi).$$

The transition kernel is the conditional distribution of tomorrow's state given today's state.

Relaxing independence preserves the basic structure of the model. The regulator still chooses the alert threshold A and issues an alert when pollution exceeds it. The consumer's prior remains unchanged. Responsiveness to an alert is still determined by the behavioral response function $\alpha(n_t)$, and n_t still tracks the recent alert history.

The difference in this case now stems from the *distribution* of recent alert histories. Under independence, the distribution of n_t only relied on the marginal alert probability determined by $F(A)$. However, under serial correlation, alert histories depend on the transition kernel G. This means that high pollution today makes high pollution tomorrow more likely, and creates a tendency for alerts to cluster over time. Accordingly, weights are now defined as

$$w_j(A,\rho;G) := \Pr(n = j \mid \phi \geq A), \qquad j = 0, 1, \ldots, \rho.$$

Now, weights depend on G, as G determines how likely an alert day is to occur after a recent alert sequence. Similarly, the distribution of pollution on alert days may itself vary across recent-alert states, as shown in Appendix A.1.3.

Relative to the i.i.d. setup, allowing for serial correlation leads to two key differences. First, changing A changes the weights $(w_j(A, \rho; G)$ both through their frequency and through how they are clustered over time. Second, the expected decision wedge now depends on the recent-alert state and can no longer be neatly summarized in a single $\mathcal{I}_1(A)$ term as in Equation 6.

This has an important implication for the optimal choice of A. Under independence, changing the threshold reallocates probability mass between alert and no-alert days, while also changing the average behavioral response through the weights on $\alpha(\cdot)$. Under serial correlation, changing A additionally changes the persistence and length of alert episodes. A lower threshold may generate longer runs of alerts, pushing more alert days into states with larger recent alert exposure, while a higher threshold may shorten alert episodes and concentrate alerts in states with lower recent alert exposure. Thus, the optimal threshold need account not only for how often alerts occur, but also for how alerts are sequenced over time and how this sequencing affects responsiveness.

3.6 Estimating Empirically

Taking the model to empirics, a number of parameters can be either estimated or directly calculated. For a given series of pollution draws, a distribution can be fit to the realized data, from which $\tilde{\phi}_0(A)$, $\mathcal{I}_0(A)$, and $\mathcal{I}_1(A)$ can be obtained. Reasonable values of ρ can be chosen to determine corresponding weights w_j. The behavioral response function $\alpha(n_t)$ can be estimated by seeing how alert responsiveness changes as a function of previous alert history. Relaxing independence, a discrete estimate of G can also be computed from the data.

4 Data

This section describes the data used in the empirical and counterfactual portions of the paper. I provide and discuss descriptive tables and figure, and then discuss how my data is matched across sources to arrive at the final analysis dataset.

4.1 Foot Traffic Data

Foot traffic data come from Advan Research, and is acquired via the Dewey Data platform. The data contain daily foot traffic counts for a variety of points-of-interest (POIs) across six U.S. cities from 2019 through 2024. These data are collected via geofencing, which uses GPS data from mobile devices to determine when a person enters a particular POI. For each POI, I observe its name and description, what North American Industry

Classification System (NAICS) code it falls under, where it is located, and how many people crossed into its boundaries on a given date.

Because I am interested in avoidance behavior in response to air quality alerts, I focus on outdoor, recreational places with low or no associated visiting costs. More specifically, I restrict the analysis to POIs in five NAICS categories: Recreational Goods Rental (532284), Historical Sites (712120), Zoos and Botanical Gardens (712130), Nature Parks and Other Similar Institutions (712190), and All Other Amusement and Recreation Industries (713990).

Table 2: Sample Composition by CBSA and NAICS Category

	Rental (1)	Historical sites (2)	Zoos/ gardens (3)	Nature parks (4)	Other recreation (5)	Total (6)	Row share (7)
Baltimore-Columbia-Towson, MD	2	37	12	586	78	715	6.4%
Boston-Cambridge-Newton, MA–NH	6	120	35	2,231	473	2,865	25.7%
Charlotte-Concord-Gastonia, NC–SC	1	35	18	331	71	456	4.1%
New York–Newark–Jersey City, NY–NJ–PA	30	195	106	3,250	929	4,510	40.5%
Philadelphia–Camden–Wilmington, PA–NJ–DE–MD	2	102	25	1,252	321	1,702	15.3%
Phoenix–Mesa–Chandler, AZ	20	19	13	682	148	882	7.9%
Total	61	508	209	8,332	2,020	11,130	
Column share	0.5%	4.6%	1.9%	74.9%	18.1%		

Note: The table reports the number of unique points of interest by CBSA and NAICS category. Row shares report each CBSA's share of all points of interest in the sample. Column shares report each NAICS category's share of all points of interest in the sample. NAICS categories are: 532284, recreational goods rental; 712120, historical sites; 712130, zoos and botanical gardens; 712190, nature parks and other similar institutions; and 713990, all other amusement and recreation industries. NAICS categories are: 532284, recreational goods rental; 712120, historical sites; 712130, zoos and botanical gardens; 712190, nature parks and other similar institutions; and 713990, all other amusement and recreation industries.

Table 2 shows the number of POIs by CBSA and NAICS category. In total, I observe 11,130 different places in my sample. About 75 percent of places fall under the nature parks category, and 40.5 percent of my sample is drawn from the New York CBSA. Table 3 shows the mean visits POIs receive by CBSA and NAICS category. Panel A show means for the overall sample, where POIs on average see 34.9 daily visits. Historical sites see the highest average number of visits, followed by zoos and botanical gardens. The New York CBSA experiences the highest average daily visits, which is consistent with it being the largest CBSA by population. Panel B shows means for the sample used in the fuzzy RD regression, which is restricted to days with forecasted AQI within 10 points of cutoff at 101. The overall mean here is 36.8 daily visits. Across CBSAs, average daily visits tend to be higher in the fuzzy RD sample, consistent with the tendency for air pollution to be higher on warmer days.

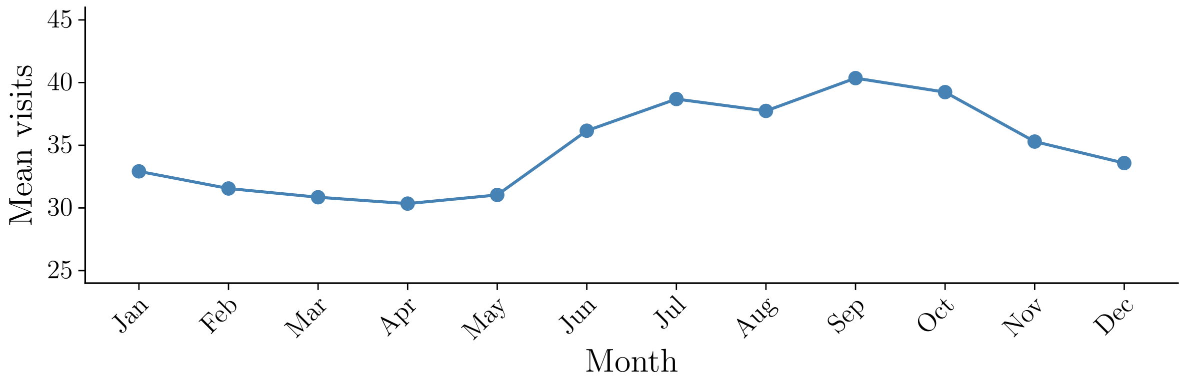


Figure 2: Seasonality in Mean Visits

Notes: This figure plots mean daily visits to outdoor recreation points-of-interest by calendar month. Each point reports the average number of visits across all observations in that month, pooling across years.

Because this dataset focuses on outdoor locations, it is unsurprising that visits exhibit a degree of seasonality. Figure 2 plots average visits across all CBSAs and years by month. Visits tend to be lower in winter months than in summer ones. This is a key consideration: If alerts occurences are distributed over the year, unadjusted comparisons may pick up seasonal differences in visits in addition to the effect of an alert. To this end, my empirical specification outlined in the next session includes a variety of time-related fixed effects in order to account for this possibility.

4.2 Air Quality Data

I obtain hourly air quality measures from the Environmental Protection Agency's Air Quality System (AQS). These data contain raw pollutant concentrations from various air quality monitoring stations across U.S. states, measured at the hourly level. I focus on two pollutants: $PM_{2.5}$ and ozone.

Working with more than one pollutant has a number of associated challenges. First, $PM_{2.5}$ and ozone are measured in different units, micrograms per cubic meter and parts per million, respectively. I take the raw pollutant concentrations and calculate their corresponding AQIs to allow comparisons of severity between them. I aggregate the AQI values to the daily level for each monitor by taking the average reading during daytime hours from 6 AM to 8 PM. Column 1 of Table 4 shows the mean actual AQI across my sample, both overall and by CBSA.

Previous work often focuses on either $PM_{2.5}$ or ozone as the pollutant of interest. However, focusing on just a single pollutant might lead to missing instances where there was in fact high pollution. Separate AQIs are calculated daily for each of the pollutants

Table 3: Mean Daily Visits per POI by CBSA and NAICS Category

	Rental (1)	Historical sites (2)	Zoos/ gardens (3)	Nature parks (4)	Other recreation (5)	Row mean (6)
Panel A: Full sample						
Overall mean = 34.9						
Baltimore-Columbia-Towson, MD	2.8	12.2	12.7	26.0	5.2	22.7
Boston-Cambridge-Newton, MA–NH	1.9	12.9	7.8	18.6	8.4	16.5
Charlotte-Concord-Gastonia, NC–SC	11.4	23.7	5.5	24.4	7.4	21.0
New York–Newark–Jersey City, NY–NJ–PA	13.8	208.6	104.6	50.9	21.8	52.8
Philadelphia–Camden–Wilmington, PA–NJ–DE–MD	1.0	69.9	11.9	34.1	15.8	32.4
Phoenix–Mesa–Chandler, AZ	4.2	16.3	17.1	32.9	10.1	27.8
Column mean	8.7	100.3	58.0	35.5	15.7	
Panel B: RD sample, bandwidth = 10						
Overall mean = 36.8						
Baltimore-Columbia-Towson, MD	3.7	13.5	13.3	28.6	5.7	25.0
Boston-Cambridge-Newton, MA–NH	2.5	13.4	7.5	20.8	9.1	18.3
Charlotte-Concord-Gastonia, NC–SC	11.7	24.5	6.8	27.8	8.5	23.7
New York–Newark–Jersey City, NY–NJ–PA	12.6	195.9	82.6	56.0	23.9	55.7
Philadelphia–Camden–Wilmington, PA–NJ–DE–MD	1.1	78.2	14.1	35.8	15.9	34.2
Phoenix–Mesa–Chandler, AZ	4.2	14.0	15.9	31.0	9.9	26.3
Column mean	8.2	97.1	47.3	38.4	16.9	

Note: The table reports mean daily visits per point of interest by CBSA and NAICS category. Means are calculated by first averaging daily visits within each point of interest, then averaging those point-of-interest-level means within each CBSA-by-NAICS category cell. Panel A uses the full analysis sample. Panel B restricts to observations within the RD bandwidth, defined as $|AQI^F - 101| \leq 10$. Row means average across points of interest within each CBSA. Column means average across points of interest within each NAICS category. NAICS categories are: 532284, recreational goods rental; 712120, historical sites; 712130, zoos and botanical gardens; 712190, nature parks and other similar institutions; and 713990, all other amusement and recreation industries.

regulated by the Clean Air Act, but the public primarily sees the maximum of those AQIs. Column 3 of Table 4 shows the breakdown of different pollutant days for my complete sample, as well as the breakdown by CBSA. Across the sample, $PM_{2.5}$ was the highest recorded AQI for about 87 percent of the days. The remaining 13 percent of days were those where ozone had the highest AQI.[12] Therefore, if I were to only focus on $PM_{2.5}$, I might falsely categorize pollution on a day to be "good" when $PM_{2.5}$ concentrations are low but ozone concentrations are high, and vice versa for ozone.

Table 4: Descriptive Statistics for AQI, Forecasts, and Alerts

	Mean AQI^A (1)	Mean AQI^F (2)	Share $PM_{2.5}$ (3)	Total Alerts (4)	Total Above (5)	Total CBSA-Days (6)	Share Alerts (7)	Share Above (8)
Panel A: Overall								
Overall	29.43	49.63	87.0%	874	943	14,594	6.0%	6.5%
Panel B: By CBSA								
Baltimore-Columbia-Towson, MD	28.82	45.58	87.1%	90	96	2,095	4.3%	4.6%
Boston-Cambridge-Newton, MA–NH	26.13	39.71	78.6%	64	65	2,024	3.2%	3.2%
Charlotte-Concord-Gastonia, NC–SC	32.71	49.61	94.9%	61	62	2,095	2.9%	3.0%
New York–Newark–Jersey City, NY–NJ–PA	28.83	48.56	86.8%	133	157	2,095	6.3%	7.5%
Philadelphia–Camden–Wilmington, PA–NJ–DE–MD	33.21	49.53	95.7%	121	125	2,095	5.8%	6.0%
Phoenix–Mesa–Chandler, AZ	26.34	68.85	78.1%	325	351	2,095	15.5%	16.8%

Note: Means are over CBSA-days. Means are calculated across all years of the data. "Share $PM_{2.5}$ " reflects the proportion of days where $PM_{2.5}$ was the highest pollutant; the remaining days are days where ozone was the highest pollutant. Days are either $PM_{2.5}$ days or ozone days by assumption, as other pollutant days are rare. "Total Alerts" reflects the sum of air quality alerts across all years, while "Total Above" shows the number of days where the forecast met or exceeded the alert cutoff of 101. "Total CBSA-Days" shows how many individual dates I have in my sample.

4.3 Air Quality Forecasts and Alert Data

In the U.S., air quality alerts are generated off of the forecasted AQI. I collect a dataset of forecasted AQI values and alert issuances from the EPA's AirNow archive. AirNow forecasts are issued at the reporting-area level, where reporting areas are regional geographic units used in AirNow products to communicate forecast and current air quality information to the public. For each reporting area, I observe the forecasted AQI value and category (Good, Moderate, Unhealthy-for-Sensitive-Groups, etc.), and whether an air quality alert was issued for a particular day.

[12]Technically, one of the other tracked pollutants (i.e., PM_{10}, carbon monoxide, sulfur dioxide, or nitrogen dioxide) could be the highest pollutant, but these instances are rare enough where focusing solely on $PM_{2.5}$ and ozone accounts for nearly all of my sample.

Column 2 of Table 4 reports the mean forecasted AQI. In my overall sample and in each CBSA, forecasted AQI is around 20 points higher than the daily actual AQI. Column 4 shows the total number of alerts that were issued across all years. Overall, days in which alerts were issued constitute about 6 percent of my sample. Column 5 displays the number of days the forecast exceeded 101 and obstensively should have triggered an alert. Note that the number of days above the threshold and the number of alerts do not align. The imperfect nature of the forecasts and fuzziness of alerts seen here echo what was seen in Figure 1 discussed previously.

Forecasted AQI exhibits seasonality. Figure 3 plots forecasted AQI over the course of my panel by CBSA. Across places, both the mean and variance of forecasted AQI increase in summer months, particularly between June and September. This seasonal pattern is consistent with atmospheric conditions that facilitate pollution formation and accumulation, as well as increased human activity during warmer months.

Figure 3 additionally shows how alerts tend to cluster. The vertical red lines drawn in the series mark dates in which an alert was issued for that CBSA. Two patterns become apparent: First, alerts tend happen in the summer, in accordance with the higher forecast, and second, alerts occur in bunches, represented by the darker and thicker bands that arise from lines being drawn close together. This is further illustrated in Figure 4, which plots the distribution of the alert history among alert days. About 23 percent of alert days have no alerts in the previous 7 days, while the remaining 77 percent of alert days have at least one alert in the previous 7 days. The serial correlation displayed in alerts is an important consideration when determining optimal policy if responsiveness fatigues.

4.4 Other Data

I obtain weather data from the Global Surface Summary of the Day (GSOD) managed by the National Oceanic and Atmospheric Administration's (NOAA) National Centers for Environmental Information (NCEI). This data contain average daily weather observations from a variety of weather stations across the U.S. I observe a variety of weather variables, including temperature, dew point, windspeed, visibility, precipitation amounts, and indicators for fog, snow, hail, and thunderstorms. While in principle these variables are not needed for my empirical strategy, in practice, alerts and forecasts could be in part be correlated with weather conditions, and so I include these data in my regressions to account for this possibility and to improve precision.

To provide evidence on the salience channel described in Section 3, I collect weekly Google Trends data for the phrase "air quality" by CBSA and year. The Google Trends measure captures relative search interest, with values normalized within each CBSA-year so that the week with the highest search volume receives a value of 100. I then match these data to the forecast and alert data to examine how public attention to air quality

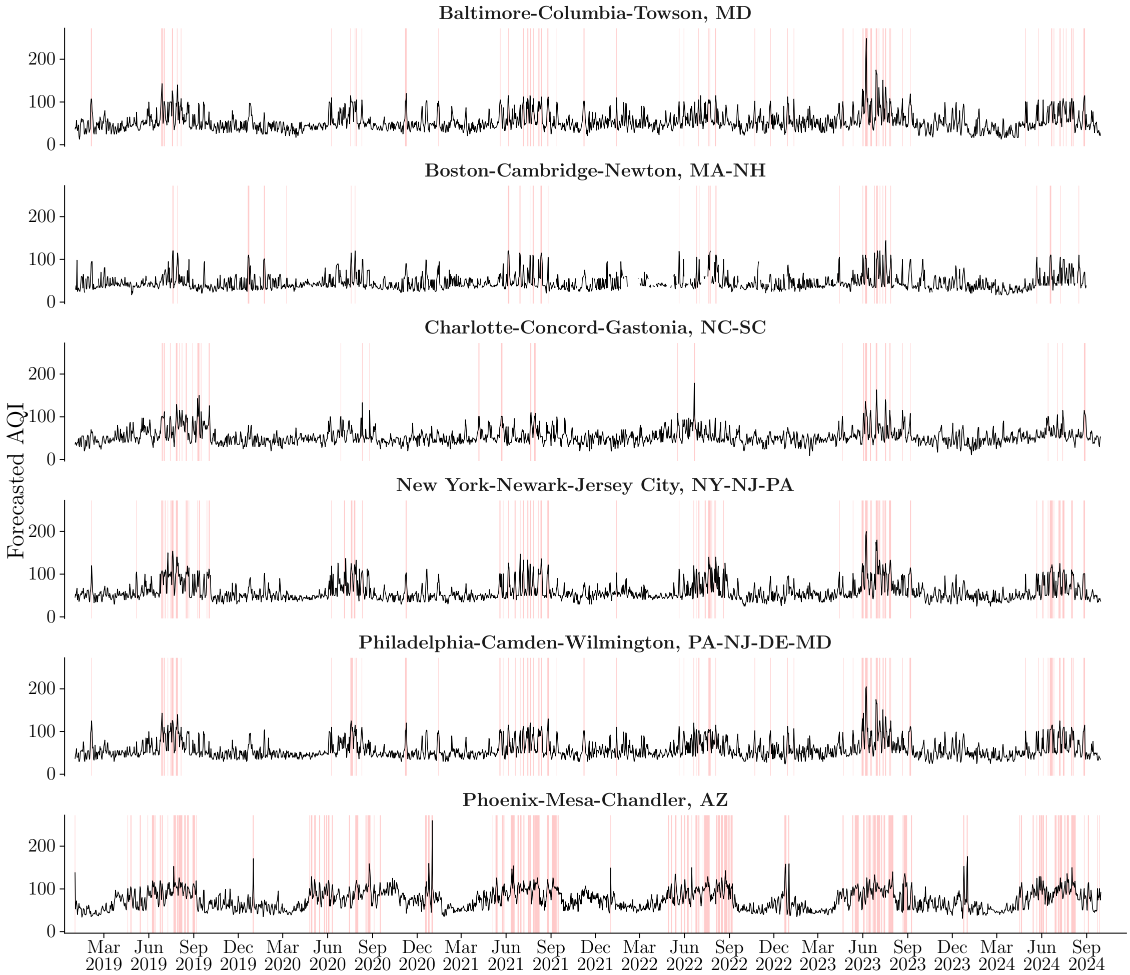


Figure 3: Serial Correlation in Forecasted AQI and the Clustering of Air Quality Alerts

Notes: The figure plots daily maximum forecasted AQI by CBSA. Each observation is collapsed to the CBSA-day level by taking the maximum forecasted AQI across observations in the analysis sample. Red vertical lines indicate days on which an air quality alert was issued.

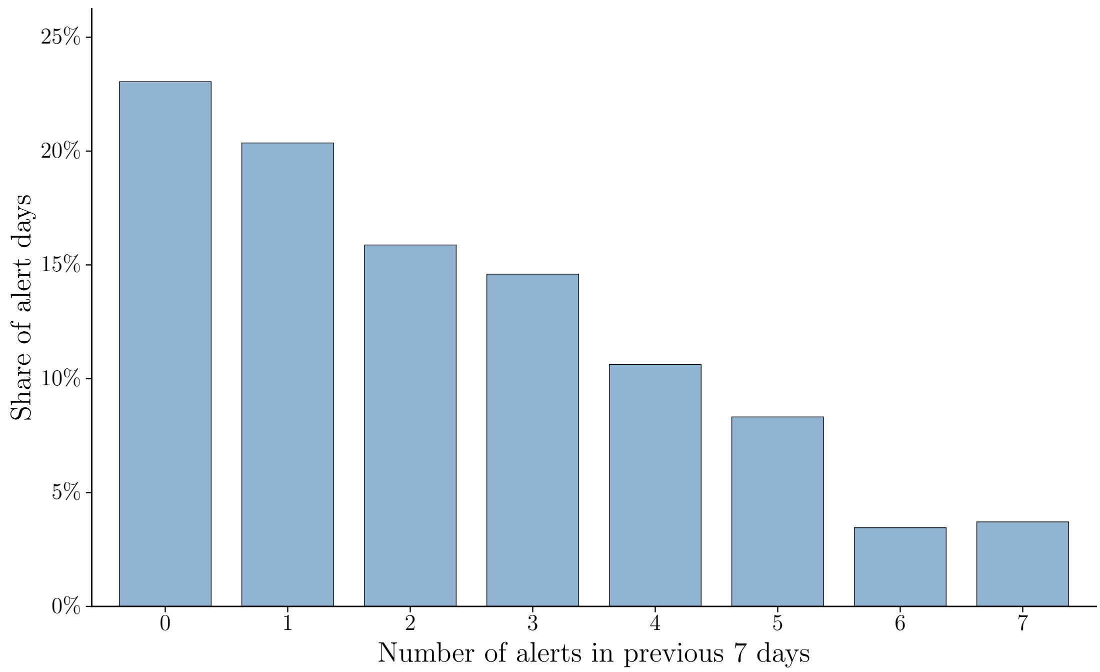


Figure 4: Distribution of Recent Alert Exposure Among Alert Days

Notes: This figure shows the distribution of recent alert exposure among alert days. The x-axis reports the number of alerts in the previous 7 days, and the y-axis reports the share of alert days in each bin.

varies with the frequency of alerts.

4.5 Dataset Construction

To merge these data together, I first match AQS data to forecasted AQI and alert data. For each monitor, I calculate the distance from the monitor to the stated coordinates of each reporting area within the same state. I then match POIs from the foot traffic data to their nearest $PM_{2.5}$ and ozone monitor. Monitors may not be operational every day, and these coverage gaps may not be random.[13] To address this, for each year, a POI is matched to a single $PM_{2.5}$ monitor and a single ozone monitor. POI-days that are not matched to both monitors are dropped from the sample. The larger of the forecasted and actual AQIs for each pollutant are taken as the POI-day's actual and forecasted AQI. A POI-day receives an alert if either its $PM_{2.5}$ forecast or its ozone forecast indicate an alert was issued. Finally, I match POI-days to the weather data by taking the closest weather station.

After matching these data sources and filtering to the appropriate bandwidth, I observe anywhere from about 470 thousand to 1.3 million POI-days across 6 CBSAs from 2019 through 2024. My data contain 11,130 unique POIs drawn from the 5 aforementioned NAICS codes. For each POI-day, I observe the number of visits to that POI, the

[13]See Zou (2021).

actual AQI for both $PM_{2.5}$ and ozone, the forecasted AQI for both $PM_{2.5}$ and ozone, whether an air quality alert was issued, and a variety of weather variables. Table 5 provides summary statistics for the overall sample as well as the sample used in the empirical design, described below.[*1]

5 Empirical Strategy

In this section, I discuss my empirical strategy. I begin by outline my fuzzy regression discontinuity (RD) model that captures the aggregate effect of an air quality alert. Then, I describe the process by which I estimate whether responsiveness changes as a function of the alert history. Finally, I conduct standard tests of the RD assumptions.

5.1 Fuzzy Regression Discontinuity

Because alert issuances are non-deterministic, some days that receive an alert where the forecast stated they should not have, and some days fail to receive an alert where the forecast indicated they should have. Resultingly, sharp RD estimates of the effect of alerts will exhibit attenuation bias. Instead, I employ a fuzzy RD design to estimate the effect of air quality alerts on behavior. More specifically, I estimate a fuzzy RD model with first stage

$$\begin{aligned} Alert_{it} &= \alpha_1\, \mathbb{1}\{AQI^F \geq 101\}_{it} + \alpha_2\,(AQI^F - 101)_{it} \\ &\quad + \alpha_3\,(\mathbb{1}\{AQI^F \geq 101\} \times (AQI^F - 101))_{it} + W_{it}'\,\psi_1 + \delta_{1i} + X_t'\,\theta_1 + \varepsilon_{it} \end{aligned} \tag{10}$$

and reduced form

$$\begin{aligned} Visits_{it} &= \beta_1\, \mathbb{1}\{AQI^F \geq 101\}_{it} + \beta_2\,(AQI^F - 101)_{it} \\ &\quad + \beta_3\,(\mathbb{1}\{AQI^F \geq 101\} \times (AQI^F - 101))_{it} + W_{it}'\,\psi_2 + \delta_{2i} + X_t'\,\theta_2 + \upsilon_{it}. \end{aligned} \tag{11}$$

Here, the term $Alert_{it}$ is an indicator variable equal to one if an alert is issued for a particular point-of-interest i on date t, while $Visits_{it}$ measures daily visits that point-of-interest. The indicator $\mathbb{1}\{\text{AQI}^F > c\}_{it}$ equals one if the forecasted AQI exceeds the cutoff $c = 101$. The term $(\text{AQI}^F - c)_{it}$ is the running variable, equal to the forecasted AQI less the cutoff. The interaction of the indicator and the running variable allows for a slope change across the cutoff. The vector W_{it} includes controls for temperature, weather, and other atmospheric conditions. The vector X_t includes year, month, weekend, and holiday fixed effects, and I also include point-of-interest fixed effects δ_i.

The fuzzy RD estimate captures the LATE and is then given by the ratio of the

[*1] **Draft note:** I've got some small discrepancies between this table and the ones above; investigating, but differences aren't large enough to worry.

Table 5: Summary Statistics for the Analysis Sample

	Full sample		RD sample	
	Mean	SD	Mean	SD
Main variables				
Visits	34.81	260.07	39.99	268.61
Forecasted AQI	51.71	20.94	100.77	4.38
Actual AQI	29.05	15.35	44.85	19.23
Air quality alert	0.033	0.178	0.494	0.500
Alerts in previous 7 days	0.23	0.76	1.23	1.56
Weather covariates				
Temperature	57.78	17.59	80.97	11.61
Dew point	43.07	17.99	59.35	14.20
Wind speed	7.08	3.45	5.72	2.15
Visibility	9.31	1.58	9.61	1.17
Precipitation	0.12	0.34	0.05	0.21
Precipitation indicator	0.390	0.488	0.269	0.444
Fog indicator	0.074	0.261	0.056	0.230
Snow indicator	0.056	0.230	0.001	0.033
Hail indicator	0.001	0.029	0.001	0.039
Thunderstorm indicator	0.070	0.254	0.185	0.388
Calendar indicators				
Weekend	0.284	0.451	0.236	0.425
Holiday	0.000	0.000	0.000	0.000
Sample dimensions				
POI-days	20,553,151		967,607	
Unique POIs	11,080		11,078	
Unique dates	2,126		674	

Note: The table reports summary statistics for the final matched POI-day dataset. The RD sample restricts observations to those within 10 AQI points of the forecasted-AQI threshold. Means and standard deviations are calculated across POI-days. Means for indicator variables can be interpreted as sample proportions. Forecasted AQI is the maximum of the $PM_{2.5}$ and ozone AQI forecasts, while actual AQI is the maximum of the corresponding realized pollutant-specific AQIs. "Alerts in previous 7 days" counts alerts issued during the seven days preceding the current observation. Temperature and dew point are measured in degrees Fahrenheit, wind speed in knots, visibility in miles, and precipitation in inches. The sample-dimension rows report counts rather than means and standard deviations.

reduced form estimate $\hat{\beta}_1$ to the first stage estimate $\hat{\alpha}_1$,

$$\hat{\gamma}_{fuzzy} = \frac{\hat{\beta}_1}{\hat{\alpha}_1}. \tag{12}$$

I estimate $\hat{\gamma}_{fuzzy}$ using two-stage least squares (2SLS).

As the fuzzy RD relies on forecasted AQI (AQI^F), I can only identify effects in places that both issue alerts and publish forecasted AQI values. While alert issuance systems are quite common among metropolitan areas, some jurisdictions choose not to include the forecasted AQI value in their reportings and instead provide just the AQI category (Good, Moderate, Unhealthy-for-Sensitive-Groups, etc.). Furthermore, forecasting methodologies may vary across CBSAs, making it difficult to back-out forecasted AQI values for places that do not publish them. My analysis thus extends only to places that both issue alerts and provide forecasted AQI values.

For clarity and consistency, I use a bandwidth of 10 AQI points on either side of the cutoff for all my RD plots and bin the running variable in 1 AQI point increments. In my tables, I also report results for a variety of different bandwidths, including the MSE-optimal bandwidth recommended by Calonico et al. (2020), calculated as 5.628 forecasted AQI points on either side of the cutoff. Due to the discrete nature of the forecast data, I report results using a bandwidth of both 5 and 6.

5.2 Estimating Response Dynamics

Equations 10 and 11 then capture the *average* contemporaneous effect of an air quality alert on visits to a place. That said, there may be response heterogeneity based on the frequency of alerts. In my theoretical framework, I discuss the role of behavioral fatigue that could lead to diminishing effects as the number of alerts increases within a given period of time. Recall the behavioral response function, $\alpha(n_t)$. The shape of $\alpha(n_t)$ is of particular policy importance, as it can suggest how frequent alerts should be issued, and thus suggest at what threshold an alert should be triggered.

To estimate the shape of the behavioral response function, I sequentially estimate Equations 10 and 11. In each iteration, I partition my sample to only include days with alerts conditional on their being n alerts in the past ρ days. When $n = 0$, my estimation captures the effect of a novel alert within the ρ time period. When $n = 1$, my estimation captures the effect of a second alert within the ρ time period, and so on. I then plot this sequence of estimates to nonparametrically estimate how the effect of an alert changes as the number of alerts increases. If estimates quickly converge to zero, this suggests that behavioral fatigue is relatively strong, and that people are less and less responsive to each subsequent alert. If estimates slowly converge to zero or stay relatively stable, this suggests that behavioral fatigue is relatively weak, and that people are still responsive to each subsequent alert. This also suggests that there could be welfare gains from lowering the alert threshold and thus issuing more alerts.

5.3 Testing RD Assumptions

Identification using a fuzzy RD in this setting requires that the running variable not be precisely manipulated around the forecasted-AQI alert threshold and that potential outcomes would evolve continuously through the cutoff in the absence of the discontinuous change in alert probability. Put differently, point-of-interest-days just below the threshold should be comparable to those just above it, except that days above the threshold are substantially more likely to receive an alert.

Because points of interest and their visitors have no control over the forecasts, any manipulation concern must arise from the forecasting agencies themselves. In particular, forecasters could potentially adjust forecasted AQI values around the alert threshold, causing days placed just above the cutoff to differ systematically from those placed just below it. However, there is no obvious general incentive for agencies to do so. First, AQI forecasting is recommended but not federally required, and forecast values do not enter federal attainment determinations (U.S. Environmental Protection Agency, Office of Inspector General, 2017). Attainment status is instead based on design values calculated from quality-assured ambient monitoring data. Second, alert issuance is discretionary rather than mechanically determined by the published forecast. An agency seeking to avoid or initiate an alert could therefore generally make that decision directly, rather than manipulate the underlying forecast.

Nevertheless, there may be narrower reasons for forecasts near the threshold to be treated differently. Agencies may wish to avoid false alarms and the reputational or administrative costs associated with unnecessary warnings, which could encourage downward adjustments. Conversely, concern about failing to warn the public on a harmful pollution day could encourage conservative upward adjustments.[14] Forecasts may also heap at focal values, such as 100 or 101, if forecasters start by forming a categorical judgment and then assign a corresponding numeric value.

Directional forecast bias does not, however, necessarily invalidate the research design. If forecasters apply such adjustments uniformly or smoothly across underlying pollution conditions, days just above and below the observed threshold may remain locally comparable. A more serious concern is instead threshold-specific sorting: forecasters might selectively place otherwise similar days on one side of the cutoff based on information that also independently predicts visits. Likewise, the tendency for forecasts to cluster on certain values is not itself sufficient to invalidate the design unless the process determining

[14]The tendency for forecasts to exceed realized AQI, shown in Figure 1, is consistent with conservative forecasting, although it does not by itself establish deliberate upward adjustment.

whether a day is assigned just above rather than just below the threshold is related to potential outcomes. These possibilities motivate the formal density and covariate-continuity tests that follow.

I first examine the distribution of the running variable in Figure 5. Figure 5a shows the full distribution of forecasted AQI, with the alert threshold at 101 marked by the dashed red line. The distribution exhibits substantial mass near the threshold. Figure 5b zooms in on the cutoff and shows that the local distribution is highly irregular, with some integer forecast values occurring far more frequently than others. This pattern is consistent with the discreteness and focal-value reporting discussed above, but the histogram alone cannot distinguish such features from threshold-specific sorting. At the same time, the visual evidence does not reveal a clear concentration of excess mass consistently on one side of the threshold rather than the other.

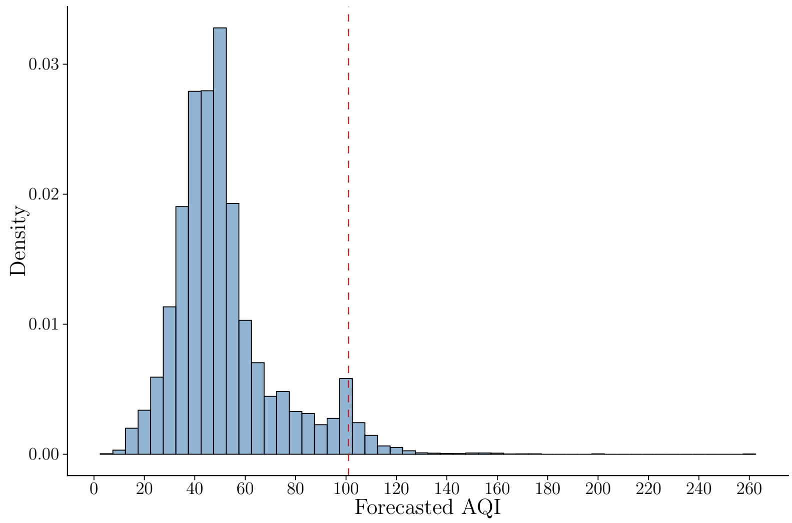
I supplement the visual evidence with formal manipulation tests. Conventional density tests reject continuity of the running-variable distribution at the threshold. The McCrary test (McCrary, 2008) yields a p-value of 0.003, while the robust Cattaneo-Jansson-Ma test (Cattaneo et al., 2020) yields a p-value below 0.001. However, these procedures are formulated around an underlying continuous density, whereas forecasted AQI is integer-valued and exhibits pronounced mass points. Conventional manipulation tests can therefore be difficult to interpret in this setting (Frandsen, 2017). I consequently also apply the Frandsen test, which is designed for discrete running variables.[15] The Frandsen test marginally rejects when the local probability mass function is constrained to be exactly linear, but it no longer rejects once even limited curvature is permitted. Taken together, the density evidence documents substantial discreteness and heaping around the cutoff, but does not provide robust evidence of precise strategic sorting.

I next examine whether observable covariates evolve smoothly through the cutoff. Figure 6 presents RD plots for the weather variables used in the analysis. Panels (a) through (e) show continuous weather measures, while panels (f) through (j) show binary weather indicators. None of the plots display a meaningful discontinuity at the threshold, indicating that observable weather conditions are similar for point-of-interest-days just below and just above the cutoff after accounting for the included fixed effects.

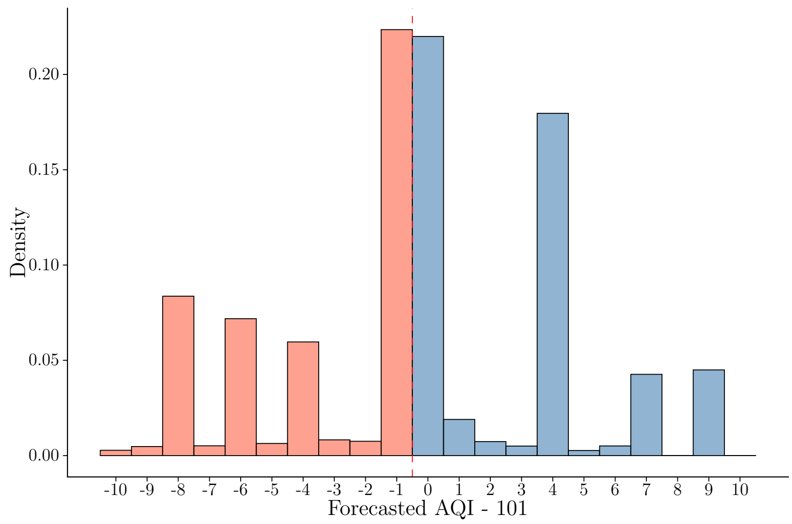
I also examine realized air quality directly. Figure 7 plots actual AQI against the forecasted-AQI running variable and likewise shows no discontinuity at the threshold. This is particularly reassuring because actual pollution is a direct determinant of outdoor activity and could confound the estimated alert effect if it changed discontinuously when the forecast crossed 101.

The fuzzy RD design also requires that crossing the forecast threshold affect visits only through its effect on alert issuance. This assumption allows visits to vary smoothly with forecasted AQI; it only rules out a separate discontinuous behavioral response at

[15] Table A.1 reports results from all three procedures.



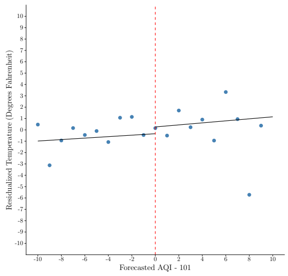
(a) Forecasted AQI distribution



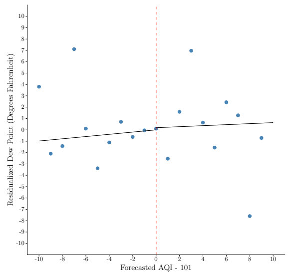
(b) Forecasted AQI relative to the alert threshold

Figure 5: Histograms of Forecasted AQI

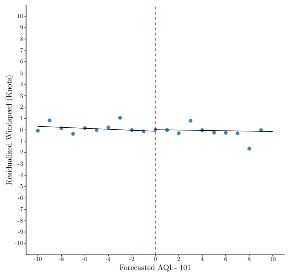
Notes: Panel (a) shows complete histogram of forecasted AQI. Panel (b) shows the histogram of forecasted AQI zoomed in on the cutoff.



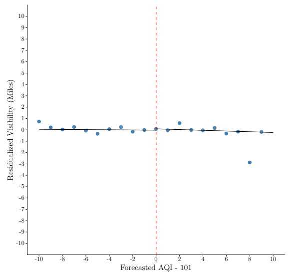
(a) Temperature



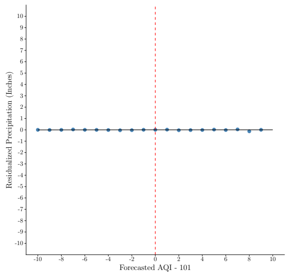
(b) Dew Point



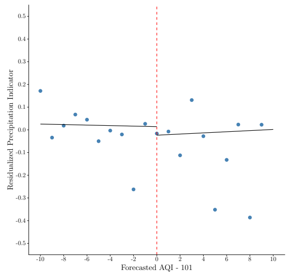
(c) Windspeed



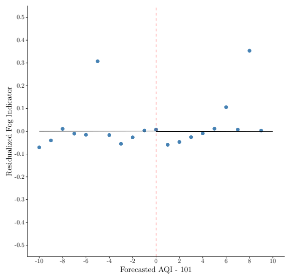
(d) Visibility



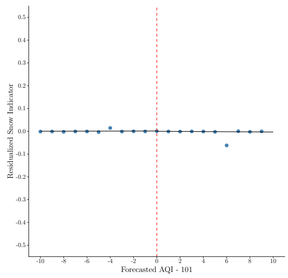
(e) Precipitation



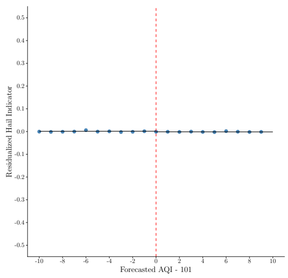
(f) Precipitation Indicator



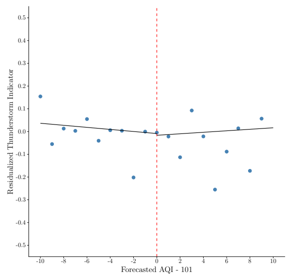
(g) Fog Indicator



(h) Snow Indicator



(i) Hail Indicator



(j) Thunderstorm Indicator

Figure 6: Residualized RD Plots for Weather Covariates

Note: Each weather covariate is residualized by place, year, month, weekend, and holiday fixed effects. Plots labeled with "Indicator" reflect covariates that are measured as binaries. The y-axis shows the units for each non-indicator covariate.

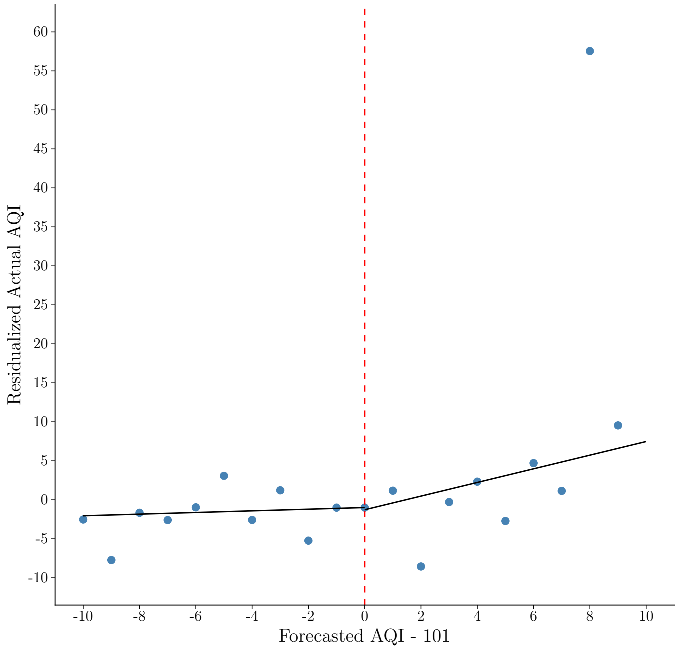


Figure 7: Residualized RD Plot for Actual AQI

Notes: This figure tests for a discontinuity in actual AQI as forecasted AQI crosses the threshold. Actual AQI is residualized by place, year, month, weekend, and holiday fixed effects.

101. Such a response would seem unlikely, since it would require individuals to observe the underlying forecast and react differently to values just above versus just below the cutoff even when no alert is issued.

6 Results

This section displays results for my empirical exercise. I begin with a discussion of my main findings for the average effect of an air quality alert on visits. I then test whether responsiveness depends on the alert history. Finally, I provide a battery of tests and robustness checks to verify the validity of my results.[*2]

6.1 Baseline Results

Figure 8 plots RD figures that show the overall effect of an alert on visits to POIs. Figure 8a shows the first stage relationship between forecasted AQI and the probability of receiving an alert, corresponding to Equation 10. The alert probability is residualized on POI, year, month, weekend, and holiday fixed effects, an on weather covariates. The size of each point reflects the number of observations that fall into each AQI value, with larger points having more observations. The first stage is strong, with the probability of receiving an alert jumping sharply after crossing the threshold by around 94 percentage points.

[*2]**Draft note:** Forthcoming based on demand.

First stage regression coefficients are reported in Panel A of Table 6, with bandwidth 10 in column (3) corresponding to the depicted RD plots. Across different bandwidths, first stage coefficients show qualitatively the same thing: crossing the threshold strongly increases the chance that an alert is issued, but with a degree on nondeterminism.

Figure 8b plots the results of reduced form regression specified in Equation 11 of the effect of crossing the threshold on visits. Visits are residualized as above. This plot shows a modest reduction in visits. Panel B of Table 6 show the fuzzy RD coefficients obtained via two-stage least squares. Across different bandwidths, days just receiving an alert see about 1.5 to 3.4 fewer visits compared to days that just miss an alert, accounting for weather covariates and the stated fixed effects. On a mean of approximately 36.8 visits, these estimates constitute about a 4 to 9 percent reduction in visits after receiving an alert. These estimates are similar with the average effect found in Graff Zivin and Neidell (2009).[16]

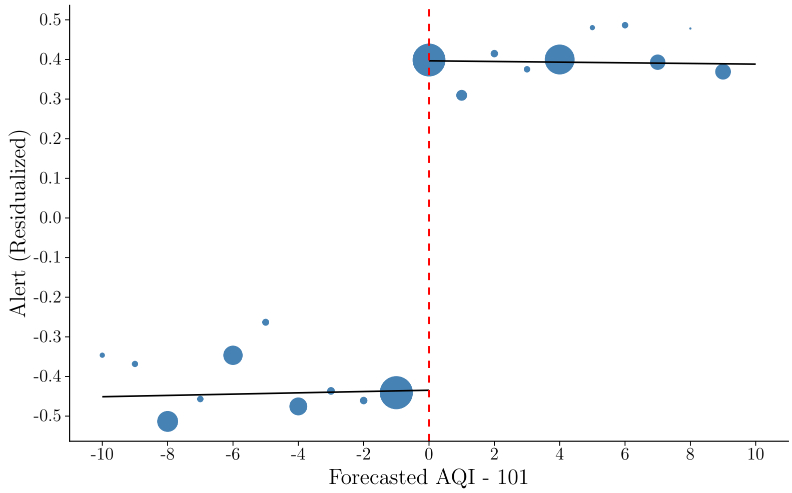
Taken together, these results point to moderate reduction in visits following an alert.

Table 6: First Stage and 2SLS Regression Results

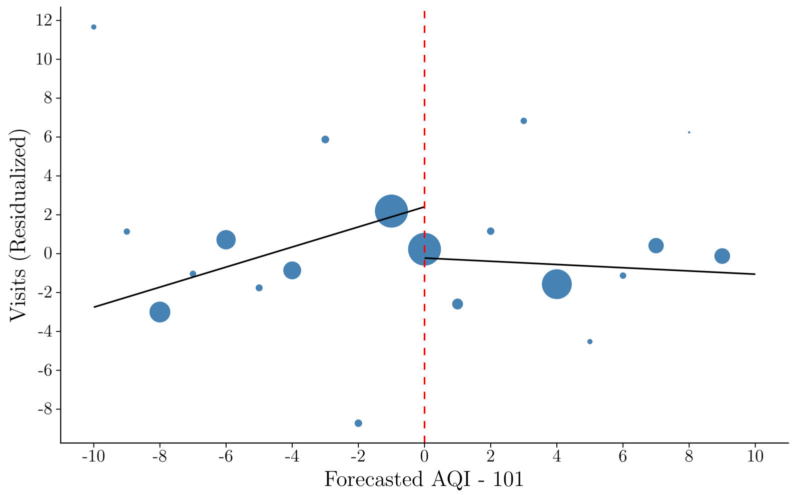
	Bandwidth				
	3	5	6	10	15
	(1)	(2)	(3)	(4)	(5)
Panel A: First stage					
$\mathbb{1}\{AQI^F \geq 101\}$	0.954^{***}	0.931^{***}	0.944^{***}	0.940^{***}	0.934^{***}
	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)
Panel B: 2SLS					
Alert	-2.573^{**}	-3.440^{***}	-2.514^{*}	-3.265^{**}	-1.518^{**}
	(0.846)	(0.745)	(1.089)	(1.130)	(0.549)
Adjusted R^2	0.578	0.578	0.591	0.584	0.597
Observations	468,389	709,756	780,060	960,856	1,288,109
Above	237,797	416,090	418,704	507,207	587,794
Below	230,592	293,666	361,356	453,649	700,315
Mean Visits	36.6	36.6	37.5	36.8	36.8

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Note: Each column reflects the first stage and reduced form RD coefficient for the corresponding bandwidth, with each being centered at 101. All regressions include point-of-interest, year, month, weekend, and holiday fixed effects, as well as weather covariates. The MSE-optimal bandwidth is 5.628, calculated using the methodology outlined in Calonico et al. (2020). Due to the discrete nature of the forecast data, results for a bandwidth of 5 and a bandwidth of 6 are reported. Standard errors are clustered at the point-of-interest level.

[16] Graff Zivin and Neidell (2009) found the average effect of an alert was about a 15 percent reduction at the Los Angeles Zoo and about a 5 percent reduction at the Griffith Park Observatory, for an average reduction of about 10 percent.



(a) First stage results



(b) Reduced form results

Figure 8: First Stage and Reduced Form RD Results

Notes: Panel (a) shows the first stage relationship between crossing the threshold and receiving an alert, while panel (b) shows the reduced form effect of crossing the threshold on visits. Both alert probability and visits are residualized on point-of-interest, year, month, weekend, and holiday fixed effects, and on weather covariates. A bandwidth of 10 AQI points is used on either side of the cutoff, which is centered at 101. The size of each point reflects the number of observations that fall into each value of the running variable.

6.2 Response Dynamics

The results above show *average* responsiveness to an air quality alert. In Section 3, I put forth a theoretical model where responsiveness to an alert depends on the alert history. It is possible there is response heterogeneity based on the number of recent alerts.

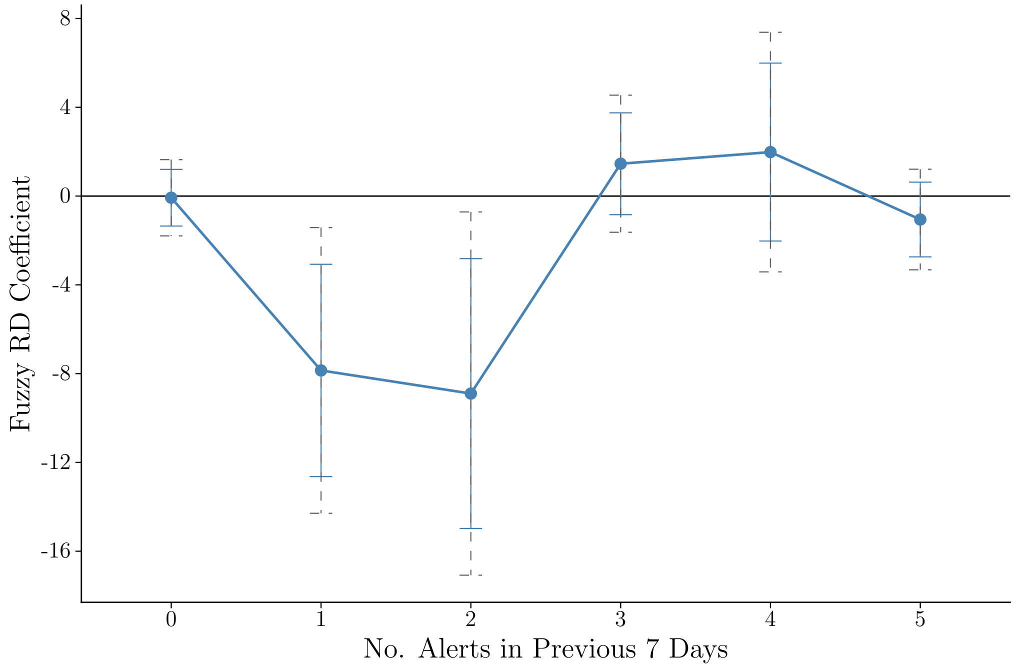


Figure 9: Alert Responsiveness Dynamics

Notes: Each point pictured here is the result of a separate fuzzy RD estimation, with the data being used conditional on the number of alerts in the previous 7 days. Each RD uses a bandwidth on 10 on either side of the cutoff. The solid blue bars reflect the 95 percent confidence interval for each estimate, with standard errors clustered at the point-of-interest level. The dashed bars represent confidence intervals with a Bonferroni adjustment for multiple hypotheses, calculated at the $1 - 0.05/6 = 0.9917$, or about 99 percent confidence intervals.

Figure 9 plots a series of fuzzy RD coefficients estimated using the process I outline in my empirical strategy in Section 5. Each point reflects the estimate from a separate fuzzy RD regression, where the x-axis shows the number of alerts in the previous 7 days that each estimate was conditioned on. Standard errors for each estimate are clustered at the POI level, shown with the solid bars, while the dashed bars reflect confidence intervals with a Bonferroni adjustment for multiple hypotheses. The estimates pictured here correspond to those in Table 7. Panel A reports the first stage relationship between crossing the threshold and receiving an alert and shows this relationship tends to strengthen as alert history builds up. Panel B reports the 2SLS coefficients for each fuzzy RD regression. I also report mean visits just below the cutoff. Visits overall tend to decrease as alert history increases, but do not go to zero.

The pattern in Figure 9 and Table 7 suggests alert responsiveness does indeed depend on the history of recent alerts. The first estimate, when there has been no alerts in the previous 7 days, captures the effect for singleton alerts, and for alerts that are the first

Table 7: Dynamic First Stage and 2SLS Regression Results

	No. Alerts in Previous 7 Days					
	0	1	2	3	4	5
Panel A: First stage						
$\mathbb{1}\{AQI^F \geq 101\}$	0.898***	0.970***	0.954***	0.994***	0.999***	1.000***
	(0.004)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
Panel B: 2SLS						
Alert	−0.075	−7.855**	−8.897**	1.456	1.982	−1.056
	(0.650)	(2.439)	(3.101)	(1.170)	(2.045)	(0.858)
Adjusted R^2	0.604	0.666	0.545	0.612	0.788	0.894
Observations	445,961	204,701	126,693	85,713	47,292	34,285
Above	202,991	107,207	73,541	57,817	26,700	26,586
Below	242,970	97,494	53,152	27,896	20,592	7,699
Mean Visits Below	44.7	53.1	49.1	33.7	39.1	35.6

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Note: Each column reflects a separate fuzzy RD estimation conditional on the number of alerts in the previous 7 days. Panel A reports the first-stage RD coefficient from regressions of alerts on an indicator for whether forecasted AQI exceeds the alert threshold. Panel B reports 2SLS estimates of the effect of alerts on visits. All regressions use a bandwidth of 10 AQI points on either side of the cutoff and include point-of-interest, year, month, weekend, and holiday fixed effects, as well as weather covariates. Standard errors are clustered at the point-of-interest level. Mean visits below the cutoff are estimated using observations below the cutoff and evaluated at the left-hand limit of the cutoff.

in a sequence. The fuzzy RD coefficient for this regression is essentially zero, meaning on average, visits do not change in response to alerts on these types of alert days. The next two estimates are negative, moderately large in magnitude, and statistically significant, showing responsiveness to alerts is strongest for the second and third alerts in a sequence. Responsiveness then diminishes for alerts where there have been 3 or more alerts in the previous 7 days.

One possible interpretation of the non-monotonic pattern seen in the coefficients is that responsiveness reflects two countervailing forces: salience and fatigue. A novel alert may arrive when air quality is not yet especially prominent, so individuals place relatively little weight on it. As alerts accumulate, repeated warnings may increase attention through media coverage, interpersonal discussion, or direct awareness of pollution conditions, particularly when pollution is visible or otherwise perceptible. At sufficiently high levels of recent alert exposure, however, fatigue may begin to offset this increase in salience. Additional alerts may then become less informative or less actionable, causing responsiveness to attenuate.

Figure 10[*3] provides descriptive evidence broadly consistent with this salience-based

[*3] **Draft note:** Because Google Trends search interest is observed only at the CBSA-week level, I

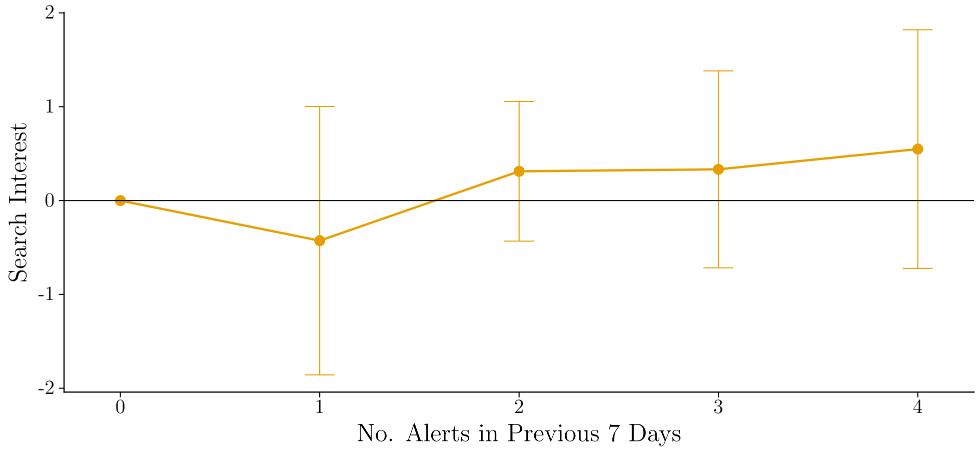


Figure 10: Search Interest by Recent Alert History

Notes: The figure plots coefficients from a regression of weekly Google Trends search interest for "air quality" on indicators for the number of alerts in the previous 7 days. Search interest is standardized within each CBSA-year. The regression includes CBSA-year fixed effects, and the omitted alert-history category is zero alerts in the previous 7 days. The sample is restricted to CBSA-weeks containing exactly one alert day, allowing each weekly search observation to be assigned to the alert-history state of that alert. Error bars show 95 percent confidence intervals based on standard errors clustered at the CBSA level. Coefficients for higher alert-history states are omitted from the figure due to lack of limited support.

interpretation. The figure plots coefficients from a regression of weekly Google Trends search interest for the term "air quality" on indicators for the number of alerts in the previous 7 days, with CBSA-year fixed effects. The point estimates suggest that search interest tends to be higher in weeks associated with larger recent-alert histories, particularly after two to four recent alerts, than in weeks where the alert follows no recent alerts. The estimates are noisy, however, and none of the plotted coefficients are statistically distinguishable from zero. The figure should therefore be viewed as suggestive evidence that attention to air quality may increase as an alert episode develops, rather than as a formal test of the salience mechanism. Taken together with Figure 9, the pattern is consistent with salience initially strengthening responsiveness before fatigue begins to reduce the marginal effect of additional alerts.

6.3 Mechanisms and Robustness Checks

6.3.1 AQI Threshold Effects

One concern with interpreting the estimates above is that behavioral responses may be driven not by alerts themselves, but by the underlying air quality threshold. The AQI scale is publicly reported and organized around discrete health-risk categories. In partic-

restrict the sample to weeks containing exactly one alert day. This allows each weekly search observation to be assigned unambiguously to the recent-alert history of that alert. The restriction is costly, however, because it excludes all weeks with multiple alert days and therefore discards a substantial share of the alert-week sample, including many of the most sustained alert episodes. Open to suggestions on other ways to do this.

ular, an AQI of 101 marks the transition from "Moderate" to "Unhealthy for Sensitive Groups." If individuals observe actual AQI values directly and respond to this category change even in the absence of an alert, then the estimated alert effects could partially reflect a behavioral response to crossing the AQI threshold itself.

To examine this possibility, I estimate a sharp RD using actual AQI as the running variable and restrict the sample to observations with no air quality alert. Specifically, I estimate

$$\begin{aligned} Visits_{it} &= \lambda_1 \mathbb{1}\{AQI^A \geq 101\}_{it} + \lambda_2 (AQI^A - 101)_{it} \\ &\quad + \lambda_3 (\mathbb{1}\{AQI^A \geq 101\} \times (AQI^A - 101))_{it} + W_{it}' \psi_3 + \delta_{3i} + X_t' \theta_3 + \epsilon_{it}. \end{aligned} \tag{13}$$

Here, AQI^A denotes actual AQI, rather than forecasted AQI. The coefficient λ_1 captures the discontinuous change in visits associated with the actual AQI crossing 101 on days without alerts. All other terms are defined as in Section 5. This exercise therefore asks whether visits change discretely when actual air quality crosses the same threshold used to define the "Unhealthy for Sensitive Groups" category, holding fixed the absence of an alert.

Figure 11 shows the corresponding RD plot. There is no visible discontinuity in visits at the actual AQI threshold. The fitted lines on either side of the cutoff connect smoothly at the cutoff. This suggests that the main estimates are unlikely to be driven by individuals responding directly to the AQI category threshold in the absence of an alert. Instead, the behavioral response appears to operate through the alert itself, rather than through a discontinuous response to actual AQI crossing 101.

7 Counterfactual Exercises

An implication of the behavioral response estimation shown in Figure 9 is that the optimal alert threshold now depends not only on the damages people incur from exposure, but also on how responsiveness changes as a function of previous alerts. Absent behavioral frictions, daily alerts would be weakly welfare increasing: alerts would have no effect on informed consumers, while misinformed consumers would update to the true pollution level. However, if repeated exposure generates fatigue, sufficiently frequent alerts may lose their efficacy. In this section, I use my data and empirical findings to simulate my theoretical model and obtain estimates of the optimal alert threshold. I focus on data from the New York-Newark-Jersey City CBSA for this exercise.

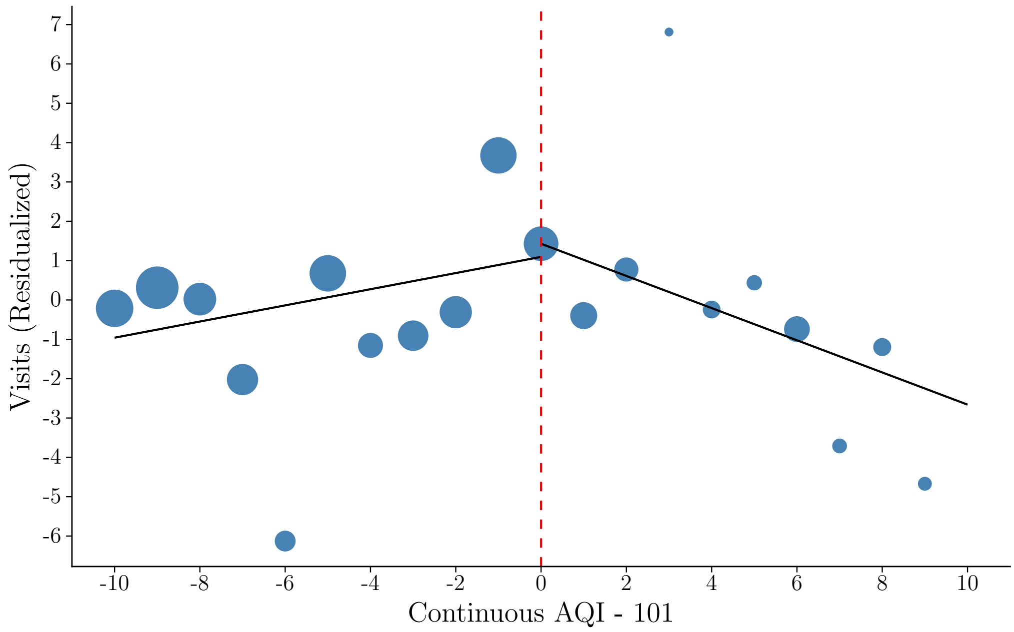


Figure 11: Sharp RD for Actual AQI in Absence of Alerts

Notes: This figure shows the sharp RD plot for visits using actual AQI as the running variable. The sample is conditioned on there being no alerts. Visits are residualized on point-of-interest, year, month, weekend, and holiday fixed effects, and on weather covariates. A bandwidth of 10 AQI points is used on either side of the cutoff, which is centered at 101. The size of each point reflects the number of observations that fall into each value of the running variable.

7.1 Setup

The theoretical model outlined in Section 3 illustrates how the optimal alert threshold balances the frequency of alerts with their effectiveness. A number of the model's parameters can be determined directly from the data. Figure 12 plots the distribution of the forecasted AQI values for the New York CBSA together with a fitted beta distribution. In the i.i.d. case, this fitted distribution is sufficient to characterize the pollution process. However, in the first-order Markov case, I additionally use the underlying AQI series to estimate transition probabilities, thereby capturing serial dependence in draws of pollution.[17]

To pin down the behavioral aspect of the model, I estimate alert responsiveness separately by recent alert history using the same iterative fuzzy RD procedure described in Section 5.2. Figure 13 plots these coefficients for the New York CBSA as a function of the number of alerts in the previous 7 days. These coefficients are then mapped to the behavioral response function, $\alpha(j)$. I assume here that $\alpha(\cdot)$ measures responsiveness relative to the most responsive alert-history state. That is, I treat the most negative

[17]The Markov process is discretized into $K = 40$ states using equal-quantile bins, with each state containing approximately $1/K$ of the observed AQI distribution. Thus, the bins have equal probability mass, although they may span unequal AQI ranges. The main simulation conclusions are qualitatively robust to alternative choices of K.

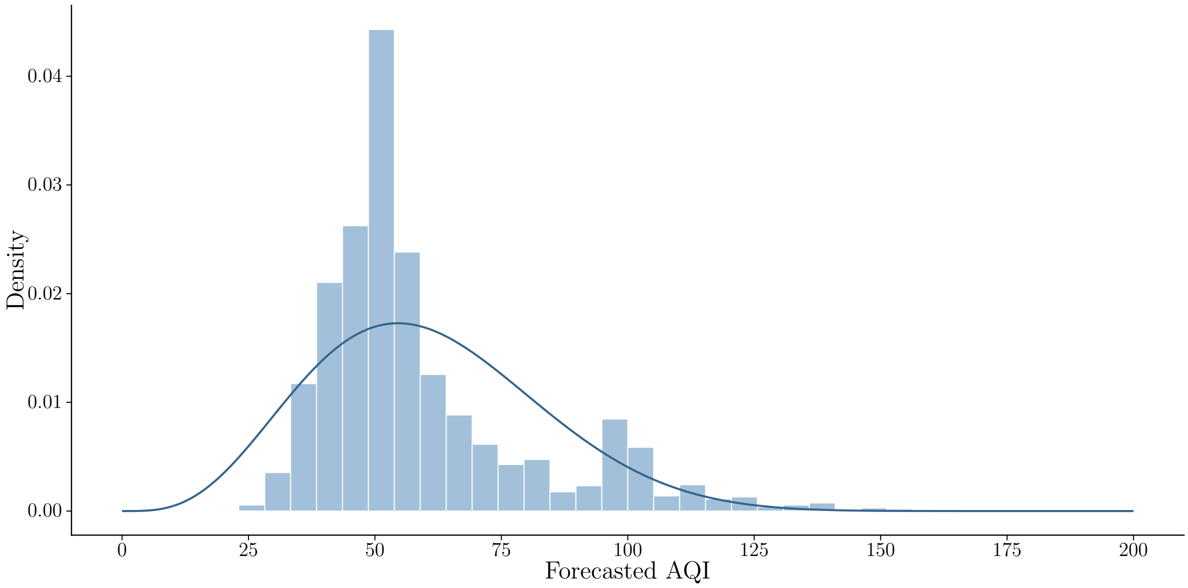


Figure 12: Distribution of Forecasted AQI for the New York CBSA

Notes: The histogram shows the distribution of daily maximum forecasted AQI in the New York-Newark-Jersey City CBSA. The solid line shows the density of the fitted scaled beta distribution used to characterize the marginal distribution of forecasted AQI in the counterfactual simulations.

coefficient as the point of maximum responsiveness, and all other negative coefficients are measured relative to this point. Statistically insignificant positive coefficients are set to zero. Additionally, for high levels of recent alert exposure, the data contain too-little support to estimate behavioral responses reliably. For these instances, I impose $\alpha(j) = 0$, that is, I assume there is no responsiveness beyond the range of the empirically supported alert-history states.

Specifically, behavioral response estimates are converted into $\alpha(\cdot)$ as follows: Let s_j denote the semi-elasticity implied by the estimated coefficient for alert-history state j. Formally,

$$s_j := -\frac{\gamma_j}{\overline{\overline{Visits_{it}}}},$$

where the mean number of visits in the New York CBSA is about 50. I normalize these semi-elasticities so that $\alpha(j)$ lies on the interval $[0, 1]$ and captures responsiveness relative to the most responsive state:

$$\alpha(j) = \frac{s_j - \min_{k\in\{0,\dots,\rho\}} s_k}{\max_{k\in\{0,\dots,\rho\}} s_k - \min_{k\in\{0,\dots,\rho\}} s_k}. \tag{14}$$

In the baseline case, the minimum semi-elasticity is zero, so this normalization reduces to scaling each s_j by the largest observed semi-elasticity. Given the estimated pollution process and behavioral response function, I evaluate expected loss for each possible threshold A. The optimal threshold, A^*, is the value of A that minimizes this objective.

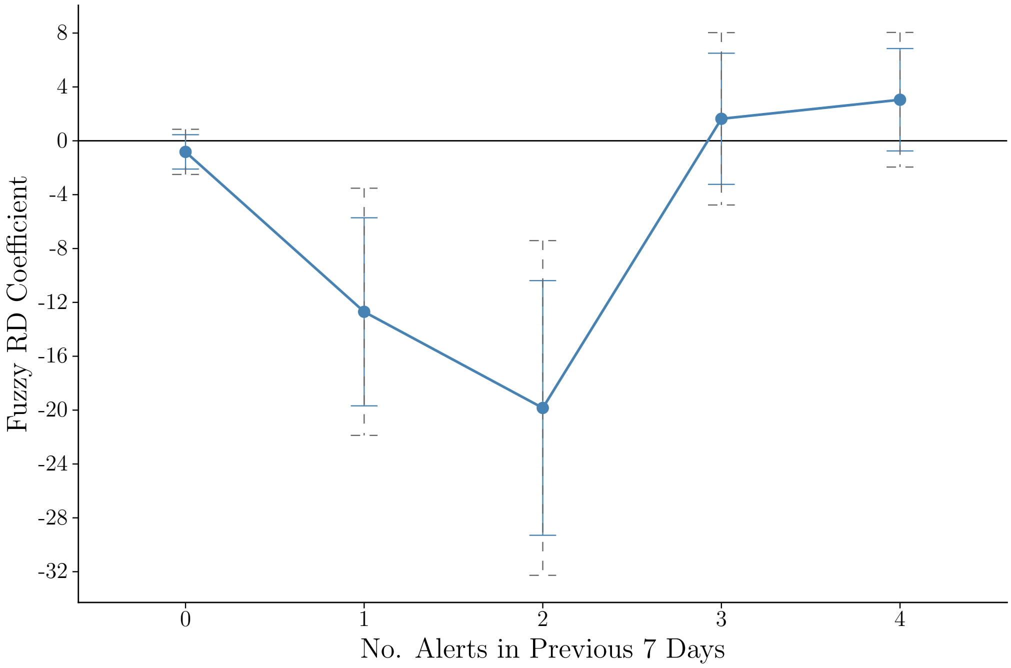


Figure 13: Alert Responsiveness Dynamics for the New York CBSA

Notes: Each point reports the estimated fuzzy RD effect of an air quality alert on visits, conditional on the number of alerts issued in the previous 7 days in the New York-Newark-Jersey City CBSA. Each estimate comes from a separate regression using a bandwidth of 10 AQI points on either side of the alert threshold. Solid error bars show 95 percent confidence intervals with standard errors clustered at the point-of-interest level, while dashed error bars show Bonferroni-adjusted confidence intervals.

7.2 Fixed Priors

In the model outlined in Section 3, I introduce a consumer whose prior is the mean level of pollution below the cutoff. Before turning to those results, I first consider a less-sophisticated consumer whose prior is fixed and does not change with the alert threshold. Figure 14 plots the resulting loss curves for fixed priors ranging from zero to 100 in 25-point increments under the Markov specification, while Table 8 reports the associated optimal thresholds, minimum losses, and alert probabilities. The figure reveals two related patterns.

First, overall loss is lower when the prior lies near the center of the forecasted AQI distribution. Because most forecasted AQI realizations in the New York CBSA are relatively moderate, priors of 50 and 75 generate smaller belief errors on most days than priors of zero, 25, or 100. The latter priors therefore produce uniformly higher loss across much of the threshold range.

Second, the loss curves are flatter when the prior is close to the center of the forecast distribution. With a prior of 50 or 75, the consumer's initial belief already approximates pollution on a large share of days. Changing the threshold therefore affects expected loss primarily on the comparatively small number of days for which realized pollution differs substantially from that prior. By contrast, when the prior is farther from typical

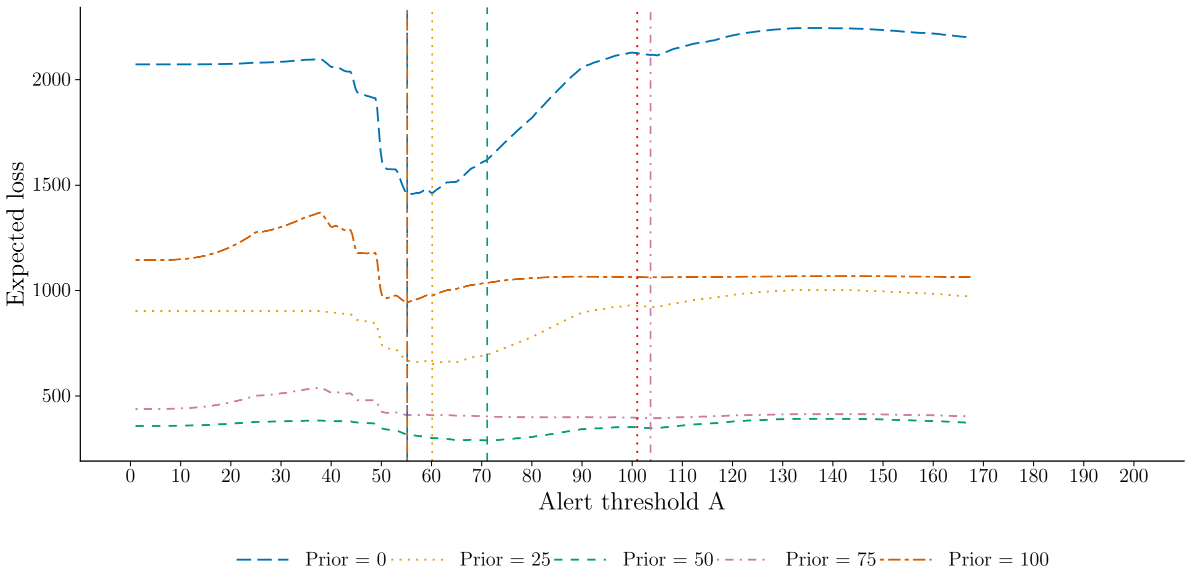


Figure 14: Expected Loss Curves Under Fixed Consumer Priors

Notes: Each line represents the expected loss curve for different fixed priors under the Markov case. Vertical lines indicate the threshold that minimizes the loss for its corresponding prior, with the exception being the dotted red line at 101, which marks the current threshold.

Table 8: Results Under Fixed Consumer Priors

	Fixed consumer prior				
	0 (1)	25 (2)	50 (3)	75 (4)	100 (5)
A^*	55.2	60.1	71.1	104.0	55.2
Loss at A^*	1,457.0	652.0	289.0	395.0	944.0
$\Pr(\phi \geq A^*)$	0.366	0.297	0.200	0.054	0.366
$W(A^*, \rho)$	0.729	0.708	0.678	0.587	0.729

Note: Each column reports the estimated optimum and associated model objects under a different fixed consumer prior. All results use the first-order Markov specification.

pollution levels, changing which days receive alerts has a larger effect on the consumer's belief error and produces greater variation in loss across thresholds.

Despite these differences in levels and curvature, the fixed-prior specifications generally favor a threshold below the current cutoff of 101.[18]

[18]The exception is the specification with a prior of 75, for which the numerical minimum lies above 101. Because the loss curve is nearly flat over this range, however, the precise location of the minimum is sensitive to small numerical and specification-dependent differences and should not be interpreted as strong evidence in favor of a higher threshold.

7.3 Baseline Results

Figure 15 plots the expected loss function over potential alert cutoffs under the i.i.d. and Markov specifications, using the variable consumer prior outlined in the model. Under the i.i.d. specification, the objective has a clearly defined minimum at approximately 77. This interior optimum reflects the central tradeoff in the model. Raising the threshold above this value leaves consumers uninformed on a larger number of moderately polluted days, while lowering it generates more frequent alerts and shifts alert days toward recent-alert states in which responsiveness is weaker.

The first-order Markov specification produces a qualitatively similar recommendation. The loss-minimizing threshold is approximately 80, but the objective is relatively flat over a broad range around the minimum. This flatness implies that the precise numerical optimum should be interpreted cautiously: nearby thresholds produce nearly identical expected losses, and relatively small changes in the estimated pollution process or behavioral-response function can shift the location of the minimum without materially changing welfare. The more robust conclusion is therefore that a range of thresholds below the current cutoff performs at least as well as, and generally better than, the status quo.

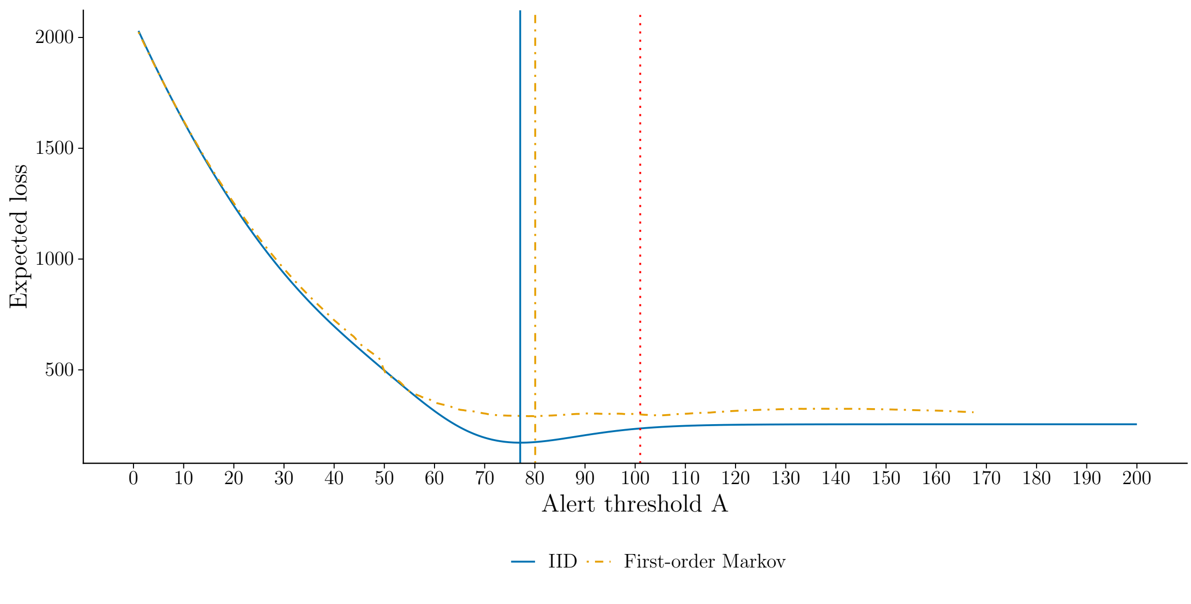


Figure 15: Baseline Expected Loss Curves

Notes: The figure plots expected loss as a function of the alert threshold under the i.i.d. and first-order Markov specifications. Vertical lines indicate the loss-minimizing threshold under each specification, while the dotted red line at 101 marks the current alert threshold.

Table 9 reports the model objects evaluated at the estimated optimum and at the current threshold of 101. Under the i.i.d. specification, the loss-minimizing threshold is 77.1. At this threshold, an alert is issued on approximately 22.7 percent of days, and the average attenuation factor is 0.393. Under the Markov specification, the estimated optimum is 80.1, corresponding to an alert probability of 15.2 percent and an average attenuation factor of 0.661. The higher attenuation factor under the Markov process is consistent with serial correlation causing alerts to cluster in recent-alert states where behavioral responsiveness is weaker.

Expected loss is lower at the estimated optimum than at the current threshold under both specifications. In the i.i.d. model, expected loss falls from 236 at the current threshold to 172 at the optimum. In the Markov model, it falls more modestly, from 301 to 291. The smaller difference in the Markov case reflects the flatness of its objective rather than evidence that the current threshold is itself optimal. A threshold near 80 minimizes the simulated objective, but the welfare penalty associated with nearby alternatives is small.

The attenuation factor does not necessarily improve monotonically as the threshold rises. In the i.i.d. model, for example, $W(A, \rho)$ is higher at the current threshold than at the estimated optimum. This occurs because high thresholds generate relatively isolated alerts and therefore place greater weight on $j = 0$, a state in which the estimated behavioral response is weak. Thus, issuing fewer alerts does not mechanically imply that the remaining alerts are more effective.

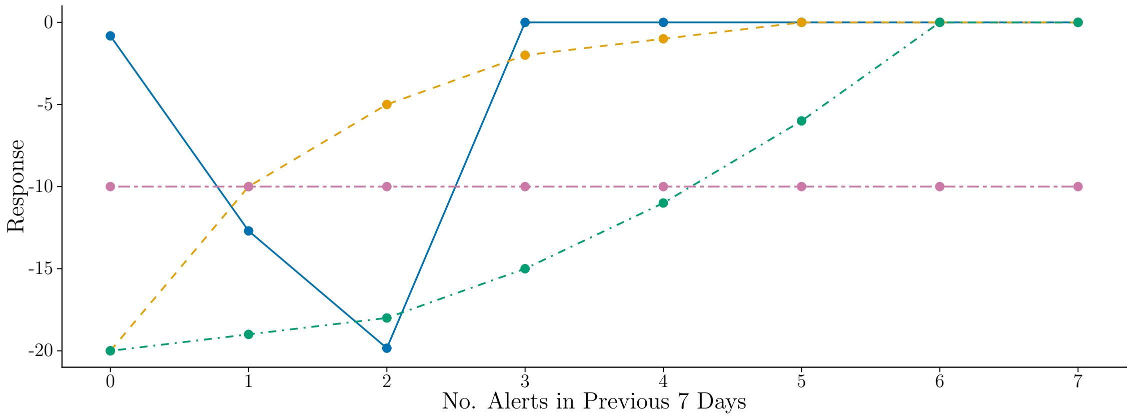
Table 9: Baseline Results versus Model Evaluation at the Current Threshold

	Optimal		Current threshold ($A = 101$)	
	IID (1)	Markov (2)	IID (3)	Markov (4)
A	77.1	80.1	101.0	101.0
Loss	172.0	291.0	236.0	301.0
$\Pr(\phi \geq A)$	0.227	0.152	0.048	0.072
$W(A, \rho)$	0.393	0.661	0.688	0.590

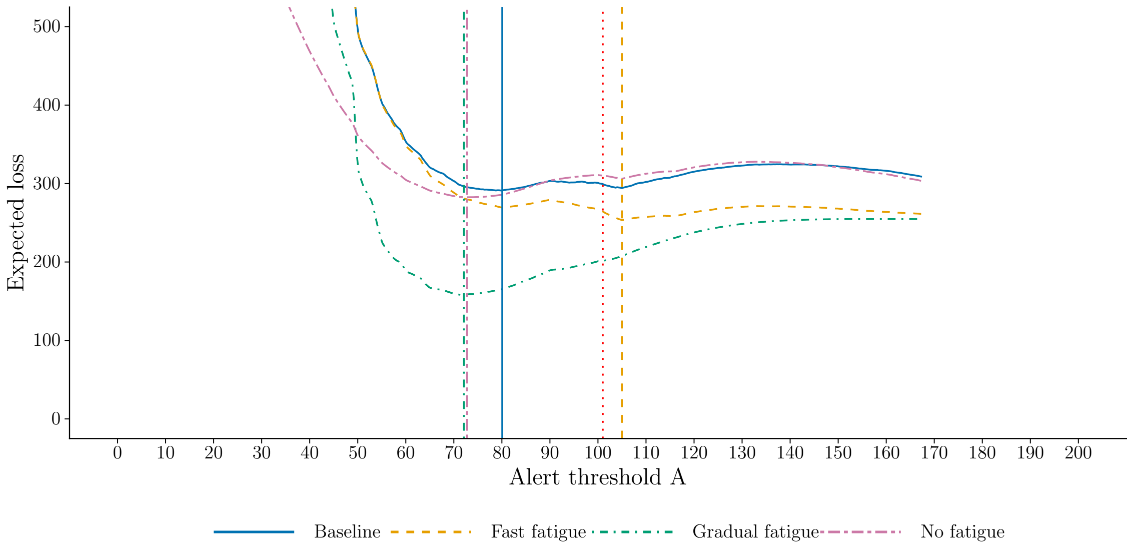
Note: The first two columns report the estimated optimum under the i.i.d. and first-order Markov models. The last two columns report the same model objects evaluated at the current threshold of $A = 101$.

7.4 Alternative Behavioral Responses

The baseline results depend in part on the shape of the behavioral-response function. I therefore examine whether the policy conclusions change under alternative assumptions about how responsiveness evolves with recent alert exposure. Figure 16a plots four response profiles. The baseline profile is constructed from the New York CBSA estimates in Figure 13. The fast-fatigue specification begins with a strong response that decays rapidly as alerts accumulate. The gradual-fatigue specification preserves responsiveness over a longer sequence of alerts, while the no-fatigue specification holds responsiveness constant across all recent-alert states.[19]



(a) Alternative Behavioral Response Curves



(b) Expected Loss Curves

Figure 16: Expected Loss Under Alternative Behavioral Response Specifications

Notes: Panel (a) shows the baseline and alternative behavioral-response functions used in the simulations, where the horizontal axis records the number of alerts in the previous 7 days. Panel (b) plots expected loss as a function of the alert threshold under each behavioral-response specification using the first-order Markov model. Colored vertical lines indicate the loss-minimizing threshold for the corresponding specification, while the dotted red line at 101 marks the current alert threshold.

Figure 16b and Table 10 report the resulting loss curves and estimated optima under the first-order Markov specification. The location of the minimum varies across response profiles, illustrating that the optimal threshold depends on how quickly alerts lose their effectiveness. The gradual-fatigue specification produces an optimum of 72.1, while the

[19]The no-fatigue case does not imply that consumers respond perfectly to alerts. It holds $\alpha(j)$ constant across recent-alert states, but at a value below one. Consumers therefore continue to place some weight on their prior even when an alert is issued. Because lowering the threshold also changes the prior $\tilde{\phi}_0(A)$, issuing alerts every day need not minimize expected loss. If instead $\alpha(j) = 1$ for all j, alerts would fully reveal pollution and an alert on every day would be weakly optimal.

no-fatigue specification produces a similar optimum of 72.8. Because alerts remain effective across a larger number of recent-alert states in these cases, the regulator can issue them more frequently without incurring as much behavioral attenuation.

By contrast, the fast-fatigue specification produces an estimated optimum of 105. When responsiveness decays quickly, lowering the threshold creates longer and more frequent alert sequences, moving a greater share of alert days into states where the warning has little effect. This model therefore favors a more selective alert policy. This result should nevertheless be interpreted in light of the shape of the objective rather than from the point estimate alone. The fast-fatigue loss curve is relatively flat near its minimum and also exhibits a local dip around the current threshold. Consequently, the estimate of 105 does not provide strong evidence that raising the threshold by four AQI points would generate a meaningful welfare gain. It instead illustrates the broader comparative-static result that more rapid fatigue shifts the preferred alert policy toward less frequent warnings.

Across the remaining specifications, the estimated optimum lies well below 101. The baseline, gradual-fatigue, and no-fatigue cases therefore support the qualitative conclusion that the current threshold is too high. At the same time, the dispersion in estimated optima—from approximately 72 to 105—shows that the exact policy recommendation is sensitive to assumptions about the pattern of behavioral responsiveness.

Table 10: Results under Alternative Behavioral Response Functions

	Baseline (1)	Fast fatigue (2)	Gradual fatigue (3)	No fatigue (4)
A^*	80.1	105.0	72.1	72.8
Loss at A^*	291.0	253.0	158.0	283.0
$\Pr(\phi \geq A^*)$	0.152	0.046	0.195	0.191
$W(A^*, \rho)$	0.661	0.306	0.223	0.640

Note: Each column reports the estimated optimum and associated model objects under an alternative specification of the behavioral-response function $\alpha(j)$. All results use the first-order Markov specification.

7.5 Improving forecasts

The preceding counterfactuals utilize forecasted AQI because alerts are triggered by forecasts rather than by realized pollution. Forecast error, however, weakens the connection between the warning and the exposure risk consumers ultimately face. To examine how this limitation affects the simulated policy problem, I compare the forecast-based model with a counterfactual in which the regulator observes perfectly predicts realized AQI.

Figure 1 shows that forecasts are only imperfectly related to realized AQI. Under perfect forecasting, the forecast for a given day would equal that day's realized AQI, so observations would lie on the 45-degree line. Instead, the data exhibit substantial dispersion, and forecasted AQI is generally higher than realized AQI.

Figure 17 presents the corresponding marginal distributions for the New York CBSA. Forecasted AQI is shifted to the right and has a heavier upper tail, including visible concentrations near 100. Realized AQI is lower on average and follows a smoother distribution. Because each series has a different empirical range, I fit its marginal distribution using its own AQI support.

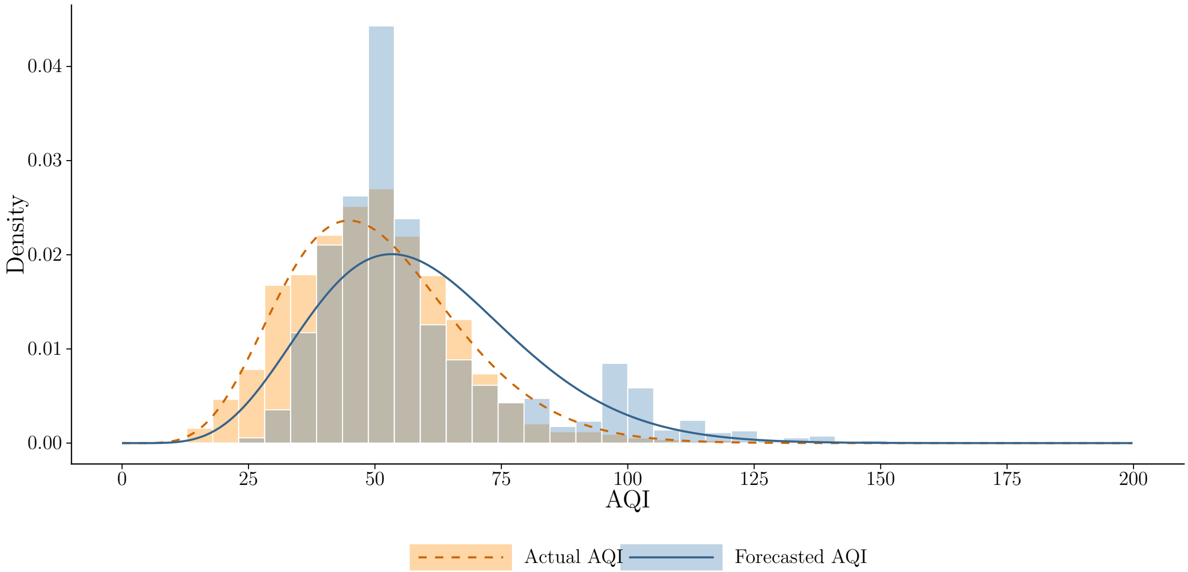


Figure 17: Distribution of Forecasted and Actual AQI for the New York CBSA

Notes: The histograms show the distributions of daily maximum forecasted and realized AQI in the New York-Newark-Jersey City CBSA. The corresponding lines show the densities of the fitted scaled beta distributions used in the counterfactual simulations. Each series is fitted using its own AQI support.

I then re-estimate the first-order Markov process using realized AQI and solve the regulator's problem under this perfect-forecasting counterfactual. This exercise changes both the marginal distribution of AQI and the transition probabilities governing its serial dynamics. It should therefore be interpreted as the combined effect of eliminating forecast error, rather than as a decomposition of any single source of forecast inaccuracy.

Figure 18 compares the two loss curves, and Table 11 reports their estimated optima. The loss-minimizing threshold is 80.1 when the model is estimated using forecasted AQI and 69.7 when it is estimated using realized AQI. In this counterfactual, eliminating forecast error shifts the preferred cutoff downward by approximately 10 AQI points.

The realized-AQI objective also has a more pronounced minimum than the forecast-based objective. The forecast-based curve remains relatively flat over a range of thresholds and contains smaller irregularities that coincide with concentrations in the forecast distribution. By contrast, the realized-AQI curve rises more clearly on either side of its minimum. This suggests that forecast quality affects not only the location of the estimated optimum, but also how sharply the policy objective identifies it.

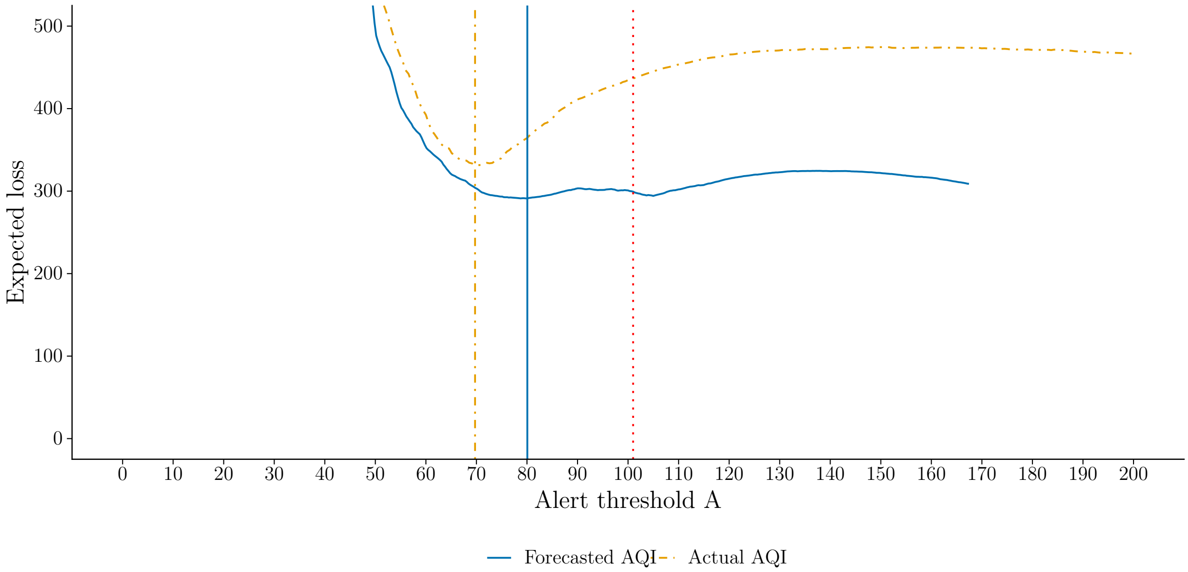


Figure 18: Expected Loss Curves for Forecasted versus Actual AQI

Notes: The figure plots expected loss as a function of the alert threshold using the distributions and transition dynamics of forecasted and realized AQI, respectively. Both curves use the first-order Markov specification. Colored vertical lines indicate the loss-minimizing threshold for the corresponding AQI series, while the dotted red line at 101 marks the current alert threshold.

Table 11: Forecast-based versus Actual-AQI Optimum

	Forecasted AQI (1)	Actual AQI (2)
A^*	80.1	69.7
Loss at A^*	291.0	331.0
$\Pr(\phi \geq A^*)$	0.152	0.092
$W(A^*, \rho)$	0.661	0.565

Note: Column (1) uses the distribution of forecasted AQI, while column (2) uses the distribution of realized AQI. All results use the first-order Markov specification.

Taken together, the counterfactual exercises provide support for a lower threshold, although the precise number depends on model assumptions and can be difficult to determine when the Markov objectives are flat. Under the baseline i.i.d. and Markov specifications, the loss-minimizing threshold lies below the current value of 101. The same conclusion holds under the baseline, gradual-fatigue, and no-fatigue response profiles, while sufficiently rapid fatigue can shift the estimated optimum upward.

The results therefore suggest that the current threshold is likely too high for the New York CBSA, while also emphasizing that the precise welfare-maximizing cutoff depends on local pollution dynamics, behavioral responsiveness, and forecast quality. The realized-AQI exercise further indicates that improvements in forecasting could complement threshold reform by aligning alerts more closely with the pollution risks consumers actually face and by making the policy objective more informative about where the threshold should be set.

8 Conclusion

Air quality alerts are one of the main ways that public agencies communicate short-run pollution risk. These alerts are low-cost, widely disseminated, and potentially valuable if they help individuals avoid exposure. However, the design of an alert system involves a behavioral tradeoff. Issuing more alerts can provide useful information on more days, but frequent warnings may also reduce the effectiveness of each additional alert. This paper studies that tradeoff and uses it to evaluate the design of air quality alert thresholds. I develop a model of optimal alert policy that incorporates history-dependent responsiveness and estimate the relevant behavioral responses using foot traffic data matched to air quality forecasts and alert issuances.

The empirical results indicate that air quality alerts reduce visits to outdoor recreational points of interest near the alert threshold. This provides evidence that public pollution information induces avoidance behavior. The response is not uniform across alert histories. Estimated effects are largest after a small number of recent alerts and then attenuate as alerts accumulate. This pattern demonstrates that the effect of an alert depends on the sequence of warnings that precedes it. It is consistent with competing behavioral forces: recent alerts may initially increase attention to air quality, while repeated exposure may eventually diminish the marginal response to additional warnings.

This history dependence changes the regulator's problem. When responses vary across prior-alert states, the average effect of an alert is not sufficient to evaluate an alternative warning policy. Changing the threshold determines not only how often alerts are issued, but also the histories under which those alerts occur. Lowering the threshold may provide information on more days while shifting a larger share of warnings into states in which recipients are less responsive. The regulator must therefore consider both the direct informational value of additional alerts and the way the policy changes their average effectiveness. This margin is especially important when pollution is serially correlated, because alerts arrive in clusters rather than as isolated events.

The counterfactual simulations suggest that the current AQI threshold of 101 may be too high, at least in the New York City CBSA. Across a range of behavioral-response specifications, the welfare-minimizing threshold generally lies below the status quo. The results support lower thresholds, but they also suggest that agencies should evaluate threshold changes using local pollution dynamics and alert histories rather than relying on a single static cutoff set at the federal level. The analysis also points to forecast quality as a key margin for improving alert policy. Alerts are based on forecasted AQI, and forecast error weakens the link between alerts and realized exposure risk. Better forecasts could allow agencies to target warnings more effectively and improve the welfare gains from threshold reform.

More broadly, this paper illustrates a general feature of policies that repeatedly provide information or prompt behavioral change. When the effect of an intervention depends on prior exposure, the policy rule helps determine the effectiveness of the interventions it generates. Evaluating a counterfactual policy therefore requires more than estimating the average response under the status quo: it requires understanding how the policy changes the frequency and sequencing of interventions and how behavior varies across the resulting histories. This consideration applies to threshold-triggered warnings for heat, wildfire smoke, severe weather, and other hazards, but also to repeated information policies more generally. The key design question is not simply how much information to provide, but how the delivery rule shapes the behavioral response to that information over time.

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A Additional Material

A.1 Derivations for the Theoretical Framework

This appendix outlines derivations of the elements used in the model in Section 3.

A.1.1 Derivation of the Decision Wedge

Here I derive the decision wedge used in Section 3. The consumer's true utility is

$$U^{TRUE}(x,\phi) = bx - \frac{1}{2}x^2 - \phi x.$$

The consumer's optimal choice under true pollution ϕ solves

$$\frac{\partial U^{TRUE}(x,\phi)}{\partial x} = b - x - \phi = 0,$$

so $x^*(\phi) = b - \phi$.

When the consumer perceives pollution to be $\tilde{\phi}_1$, perceived utility is

$$U^{PERC}(x,\tilde{\phi}_1) = bx - \frac{1}{2}x^2 - \tilde{\phi}_1 x.$$

The perceived optimal choice solves

$$\frac{\partial U^{PERC}(x,\tilde{\phi}_1)}{\partial x} = b - x - \tilde{\phi}_1 = 0,$$

so $\tilde{x}(\tilde{\phi}_1) = b - \tilde{\phi}_1$.

The decision wedge is the difference between true utility evaluated at the true optimal choice and true utility evaluated at the choice induced by perceived pollution:

$$\Delta(\phi,\tilde{\phi}_1) = U^{TRUE}(x^*(\phi),\phi) - U^{TRUE}(\tilde{x}(\tilde{\phi}_1),\phi).$$

First, true utility evaluated at $x^*(\phi) = b - \phi$ is

$$\begin{aligned}
U^{TRUE}(x^*(\phi),\phi) &= b(b-\phi) - \frac{1}{2}(b-\phi)^2 - \phi(b-\phi) \\
&= b^2 - b\phi - \frac{1}{2}(b^2 - 2b\phi + \phi^2) - b\phi + \phi^2 \\
&= \frac{1}{2}b^2 - b\phi + \frac{1}{2}\phi^2.
\end{aligned}$$

Next, true utility evaluated at $\tilde{x}(\tilde{\phi}_1) = b - \tilde{\phi}_1$ is

$$\begin{aligned}
U^{TRUE}(\tilde{x}(\tilde{\phi}_1), \phi) &= b(b - \tilde{\phi}_1) - \frac{1}{2}(b - \tilde{\phi}_1)^2 - \phi(b - \tilde{\phi}_1) \\
&= b^2 - b\tilde{\phi}_1 - \frac{1}{2}(b^2 - 2b\tilde{\phi}_1 + \tilde{\phi}_1^2) - b\phi + \phi\tilde{\phi}_1 \\
&= \frac{1}{2}b^2 - b\phi - \frac{1}{2}\tilde{\phi}_1^2 + \phi\tilde{\phi}_1.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\Delta(\phi, \tilde{\phi}_1) &= \left(\frac{1}{2}b^2 - b\phi + \frac{1}{2}\phi^2\right) - \left(\frac{1}{2}b^2 - b\phi - \frac{1}{2}\tilde{\phi}_1^2 + \phi\tilde{\phi}_1\right) \\
&= \frac{1}{2}\phi^2 + \frac{1}{2}\tilde{\phi}_1^2 - \phi\tilde{\phi}_1 \\
&= \frac{1}{2}(\phi - \tilde{\phi}_1)^2.
\end{aligned}$$

Thus, under the assumed quadratic utility function, the welfare loss from misperceiving pollution is exactly proportional to the squared difference between true and perceived pollution:

$$\Delta(\phi, \tilde{\phi}_1) = \frac{1}{2}(\phi - \tilde{\phi}_1)^2.$$

A.1.2 Derivation of the Expected Decision Wedge

The decision wedge $\Delta(\cdot)$ can be broken into two cases based on whether an alert is issued or not. When there is not an alert, $\phi_t < A$, and $\tilde{\phi}_{1,t} = \tilde{\phi}_0$. So,

$$\delta_{0,t}(\phi_t) := \Delta_t(\phi_t, \tilde{\phi}_0) = \frac{1}{2}(\phi_t - \tilde{\phi}_0(A))^2. \tag{A.1}$$

In words, the decision wedge for a day t that does not receive an alert is proportional to the squared difference between true pollution ϕ_t and the consumer's prior perceived pollution level. Call this difference the error.

When there is an alert, $\phi_t \geq A$, and $\tilde{\phi}_{1,t} = (1 - \alpha(n_t))\tilde{\phi}_0(A) + \alpha(n_t)\phi_t$. Then, $\phi_t - \tilde{\phi}_1 = (1 - \alpha(n_t))(\phi_t - \tilde{\phi}_0(A))$, so

$$\delta_{1,t}(\phi_t, \alpha(n_t)) := \Delta(\phi_t, \tilde{\phi}_{1,t}(\phi_t, \alpha(n_t))) = \frac{1}{2}(1 - \alpha(n_t))^2(\phi_t - \tilde{\phi}_0(A))^2, \tag{A.2}$$

that is, the decision wedge for a day t that does receive an alert is proportional to the squared difference between true pollution ϕ_t and the consumer's prior perceived pollution level, scaled by $(1 - \alpha(n_t))^2$. When $\alpha(n_t)$ is large, $(1 - \alpha(n_t))^2$ will be relatively close to zero, which will shrink the squared error. Consider the extreme case when $\alpha(n_t) = 1$. Then, the error will be zero, since all the weight in the first case of Equation 3 is placed on ϕ_t.

Now, using the law of total expectation over the two cases $\phi_t < A$ and $\phi_t \geq A$, the expected decision wedge for a day t can be written as

$$\mathbb{E}[\Delta_t] = \Pr(\phi_t < A)\mathbb{E}[\delta_{0,t}(\phi_t) \mid \phi_t < A] + \Pr(\phi_t \geq A)\mathbb{E}[\delta_{1,t}(\phi_t, n_t) \mid \phi_t \geq A]. \tag{A.3}$$

This expression shows that expected welfare loss is a probability-weighted average of the decision wedge in the no-alert and alert cases. The threshold A affects this object both by changing the probability of each case and by changing the wedge realized within each case. For a given threshold A, I assume the pollution process is stationary, and since the alert rule is time-invariant, the induced process for alerts, alert histories, and decision wedges is stationary as well. Accordingly, time subscripts t can be suppressed and Equation A.3 can be rewritten as

$$\mathbb{E}[\Delta] = \Pr(\phi < A)\mathbb{E}[\delta_0(\phi) \mid \phi < A] + \Pr(\phi \geq A)\mathbb{E}[\delta_1(\phi, n) \mid \phi \geq A]. \tag{4}$$

On alert days, responsiveness to alerts depends on the recent-alert state n. For a given threshold A, define weights

$$w_j(A, \rho) := \Pr(n = j \mid \phi \geq A), \qquad j = 0, 1, \ldots, \rho.$$

That is, $w_j(A, \rho)$ is the proportion of alert days that occur when there have been j alerts in the previous ρ days. Then,

$$\mathbb{E}[\delta_1(\phi, n) \mid \phi \geq A] = \sum_{j=0}^{\rho} w_j(A, \rho)\, \mathbb{E}[\delta_1(\phi, j) \mid \phi \geq A, n = j].$$

Among alert days, the expected wedge depends on how alert days are distributed across recent-alert states. The overall expected alert-day wedge is therefore the average of the wedge in each state j, weighted by the share of alert days that occur in that state.

Now, because ϕ has cdf F and pdf f, $\Pr(\phi < A) = F(A)$ and $\Pr(\phi \geq A) = 1 - F(A)$, define conditional means for δ_0 and δ_1 as

$$\bar{\delta}_0(A) := \mathbb{E}[\delta_0(\phi) \mid \phi < A] = \frac{1}{F(A)} \int_0^A \frac{1}{2}\big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi \tag{A.4}$$

and

$$\bar{\delta}_1(A, j) := \mathbb{E}[\delta_1(\phi, j) \mid \phi \geq A] = \frac{1}{1 - F(A)} \int_A^1 \frac{1}{2}(1 - \alpha(j))^2 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi. \tag{A.5}$$

Therefore, Equation 4 can be rewritten as

$$\mathbb{E}[\Delta] = F(A)\bar{\delta}_0(A) + \big(1 - F(A)\big) \sum_{j=0}^{\rho} w_j(A,\rho)\bar{\delta}_1(A,j). \tag{A.6}$$

Then, substituting in $\bar{\delta}_0(A)$ and $\bar{\delta}_1(A,j)$ and simplifying:

$$\begin{aligned}
\mathbb{E}[\Delta] = F(A)&\frac{1}{F(A)} \int_0^A \frac{1}{2}\big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi \\
&+ \big(1 - F(A)\big) \sum_{j=0}^{\rho} w_j(A,\rho) \frac{1}{1 - F(A)} \int_A^1 \frac{1}{2}(1 - \alpha(j))^2 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi \\
&= \frac{1}{2}\left[\int_0^A \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi + \sum_{j=0}^{\rho} w_j(A,\rho) \int_A^1 (1 - \alpha(j))^2 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi\right].
\end{aligned}$$

Note that $\alpha(\cdot)$ does not depend on ϕ. It can thus be pulled out of the integral to arrive at

$$\begin{aligned}
\mathbb{E}[\Delta] = \frac{1}{2}\Big[&\int_0^A \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi \\
&+ \sum_{j=0}^{\rho} w_j(A,\rho)(1 - \alpha(j))^2 \int_A^1 \big(\phi - \tilde{\phi}_0(A)\big)^2 f(\phi)\, d\phi\Big].
\end{aligned} \tag{A.7}$$

A.1.3 Derivation of the Expected Decision Wedge with Serial Correlation

The distribution of pollution on alert days may itself vary across recent-alert states. Let the decision wedge for day t be defined as in Equation A.2. On a given day with pollution ϕ_t and state n_t, there is no difference in the calculation of the wedge. There is a difference, however, in the distribution of the wedge. Define the state-specific alert-day conditional mean as

$$\begin{aligned}
\bar{\delta}_{1,j}(A, j; G) &:= \mathbb{E}[\delta_1(\phi_t, \alpha(j)) \mid \phi_t \geq A,\ n_t = j] \\
&= \frac{1}{2}(1 - \alpha(j))^2\, \mathbb{E}\left[\Big(\phi_t - \tilde{\phi}_0(A)\Big)^2 \mid \phi_t \geq A,\ n_t = j\right].
\end{aligned}$$

Then, an analog to Equation A.6 can be written as

$$\mathbb{E}[\Delta] = F(A)\bar{\delta}_0(A) + \big(1 - F(A)\big) \sum_{j=0}^{\rho} w_j(A,\rho;G)\bar{\delta}_{1,j}(A,j;G)$$

using the same $\bar{\delta}_0(A)$ as defined in Equation A.4. Let

$$\mathcal{I}_{1,j}(A;G) := \mathbb{E}\left[\big(\phi_t - \tilde{\phi}_0(A)\big)^2 \mid \phi_t \geq A,\ n = j\right].$$

Then, after substituting and simplifying, the regulator's problem becomes

$$\min_{A} \ \mathbb{E}[\Delta] = \frac{1}{2}\left[\mathcal{I}_0(A) + \big(1 - F(A)\big)\sum_{j=0}^{\rho} w_j(A,\rho;G)(1-\alpha(j))^2\mathcal{I}_{1,j}(A;G)\right], \tag{A.8}$$

where the $\mathcal{I}_0(A)$ term is defined the same as in Equation 5.

A.2 Density Test Results

Table A.1: Manipulation Tests for Forecasted AQI

Test	Specification	p-value
McCrary (2008)	Automatic bin width and bandwidth	0.003
Cattaneo, Jansson, and Ma	Robust bias-corrected	<0.001
Frandsen (2017)	$K = 0$	0.043
	$K = 0.005$	0.047
	$K = 0.010$	0.056
	$K = 0.025$	0.122
	$K = 0.050$	0.360
	$K = 0.100$	0.890

Notes: The table reports manipulation tests for forecasted AQI at the alert threshold of 101. The McCrary test reports a test of continuity in the log density of the running variable. The Cattaneo-Jansson-Ma row reports the robust bias-corrected density-discontinuity test with adjustment for mass points. Both procedures approximate the running-variable distribution using continuous densities, whereas forecasted AQI is integer-valued and exhibits substantial mass points.

The Frandsen test is designed for a discrete running variable. The parameter K places an upper bound on the curvature of the local probability mass function. When $K = 0$, the probability mass function is restricted to be locally linear across the support points immediately surrounding the cutoff. Larger values of K allow progressively greater local curvature. The support points used by the Frandsen test are forecasted AQI values of 100, 101, and 102, which contain 3,032, 1,776, and 314 observations, respectively.

A.3 Additional Figures

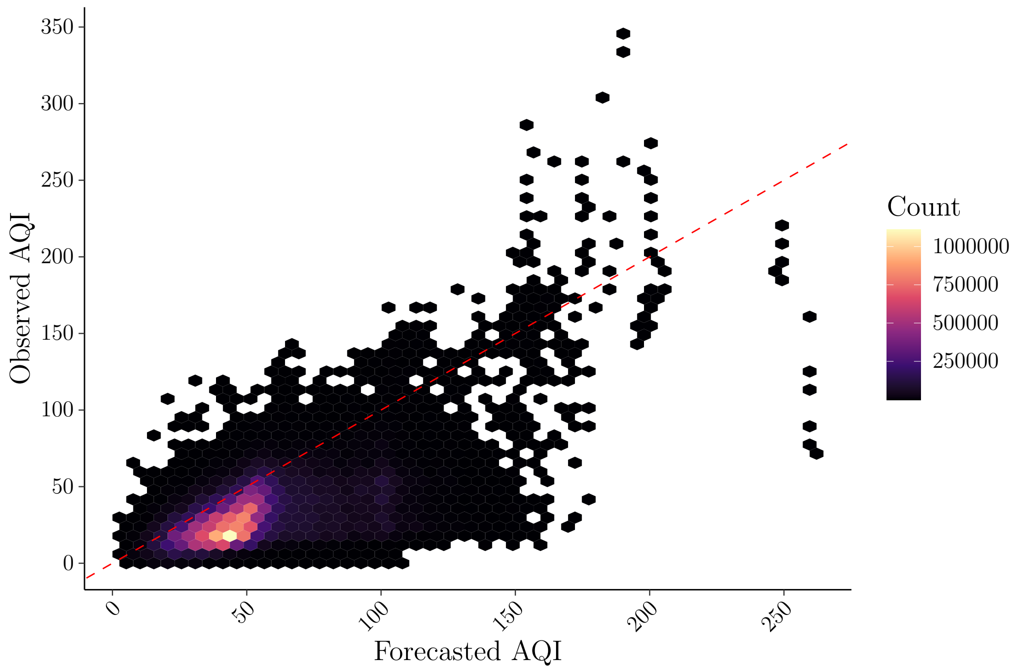

Figure A.1: Forecast Accuracy with Frequency

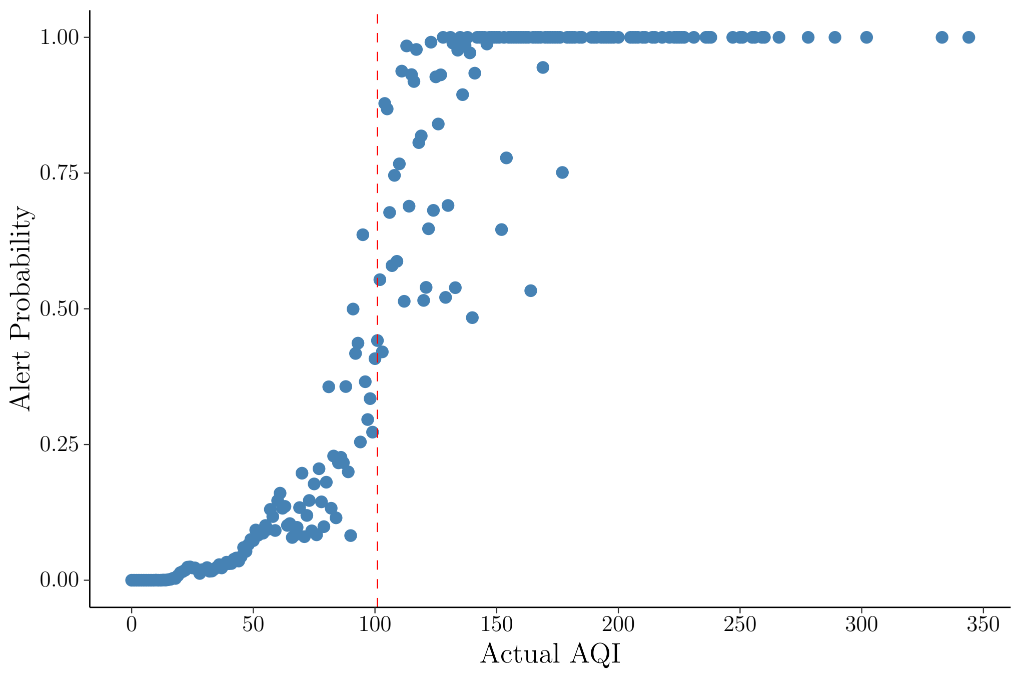

Figure A.2: Actual AQI vs. Alert Probability

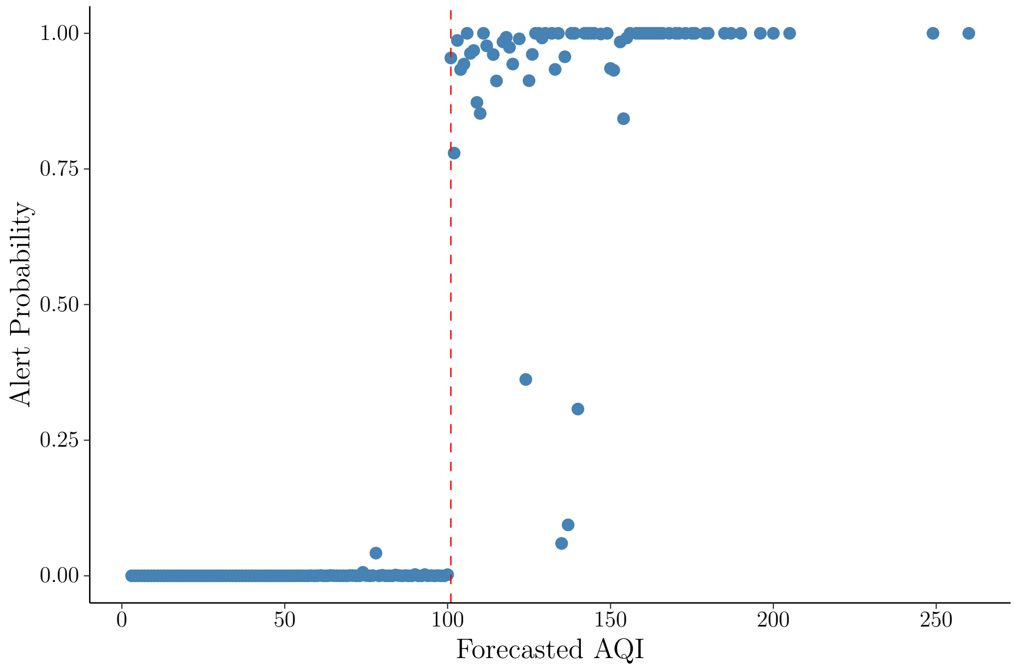

Figure A.3: Forecasted AQI vs. Alert Probability